

EEE 263 – ELECTRONIC CIRCUITS

Diode

Khairul Islam

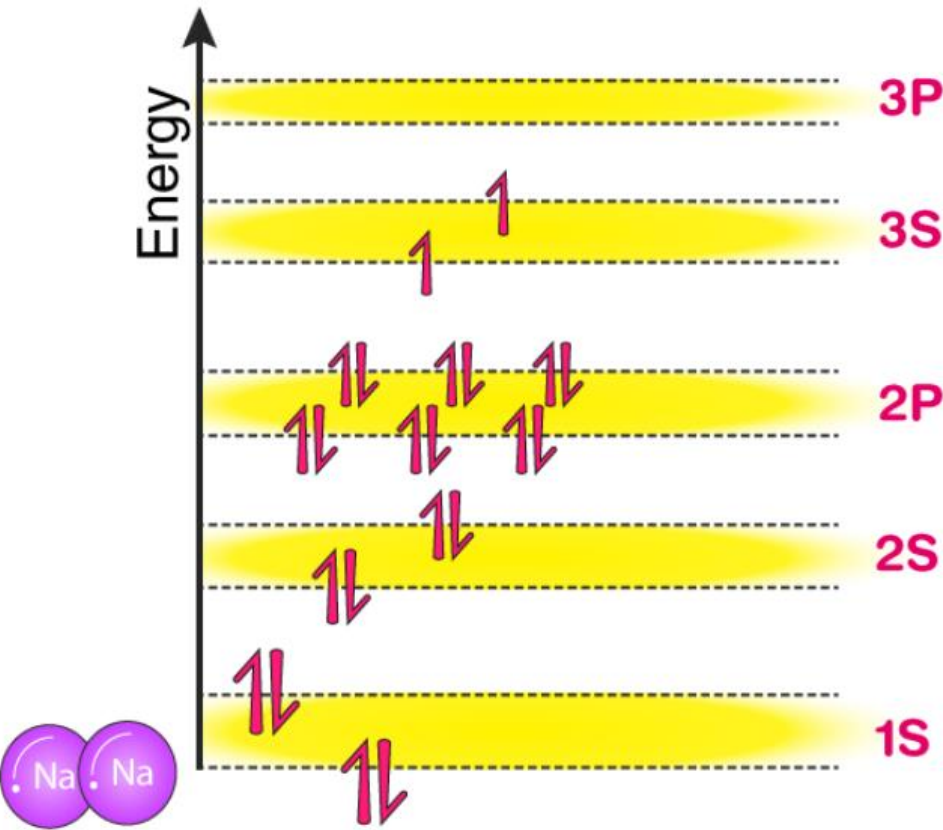
Lecturer

Department of EEE, BUET

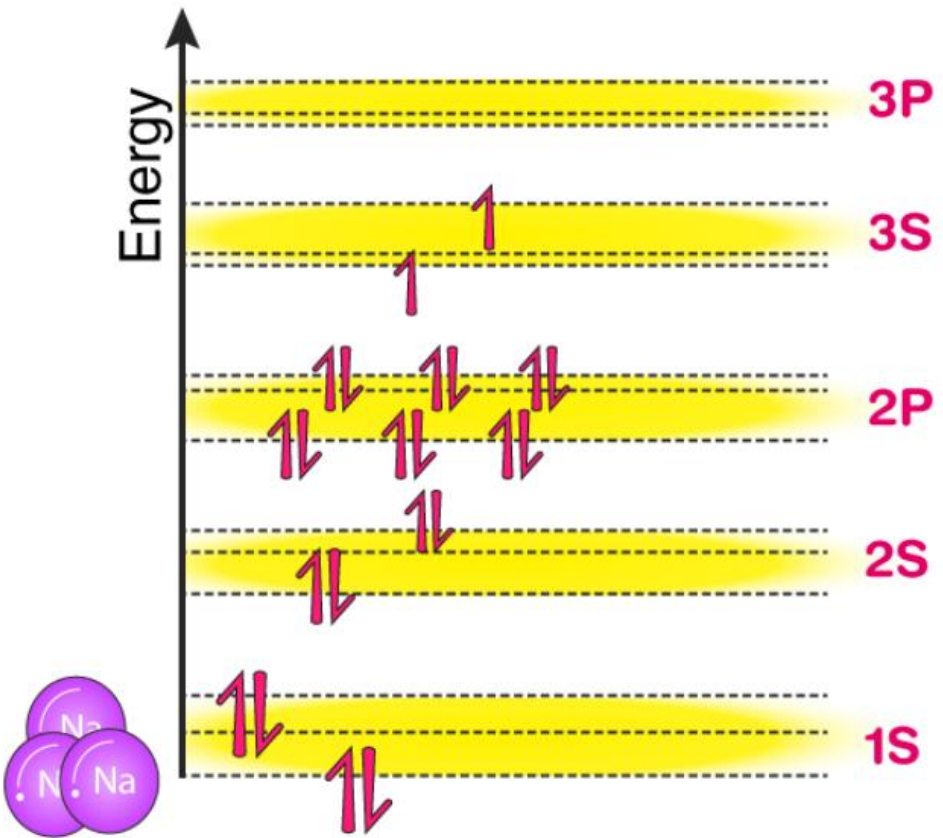
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Band Theory

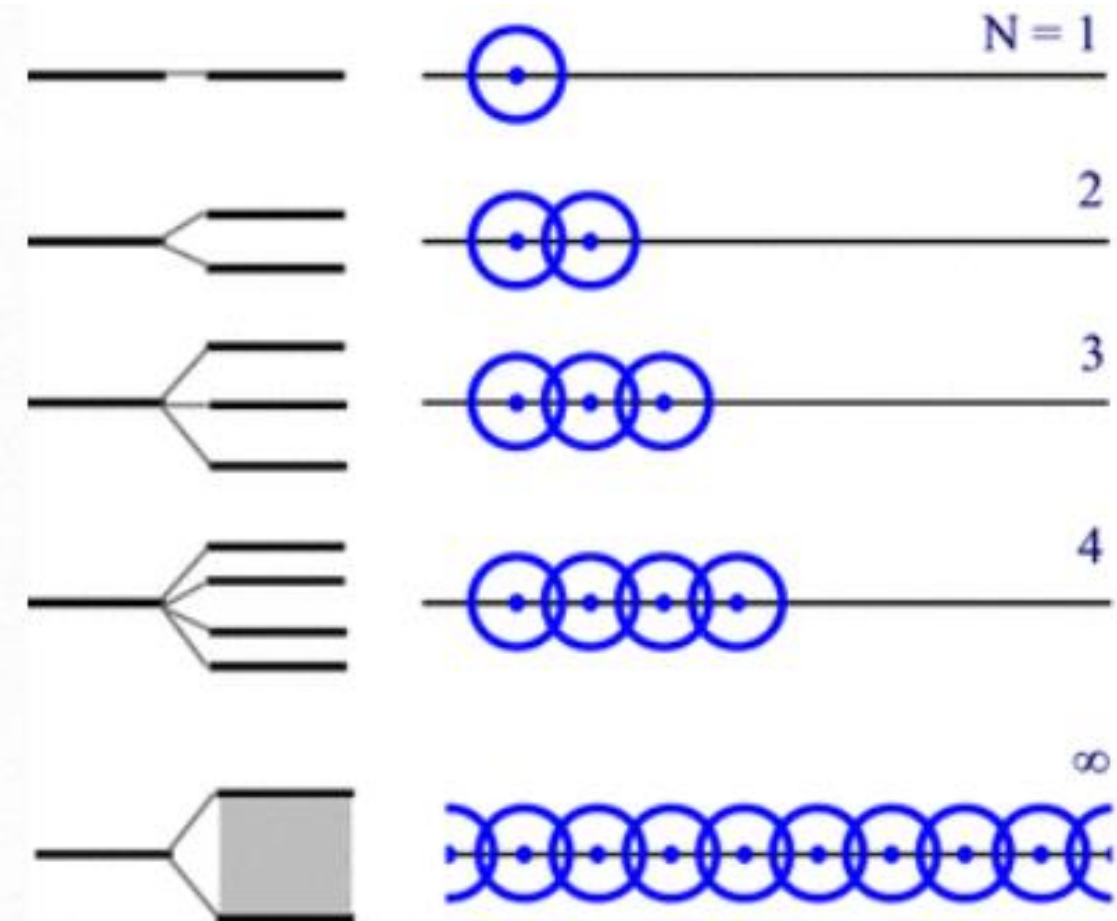
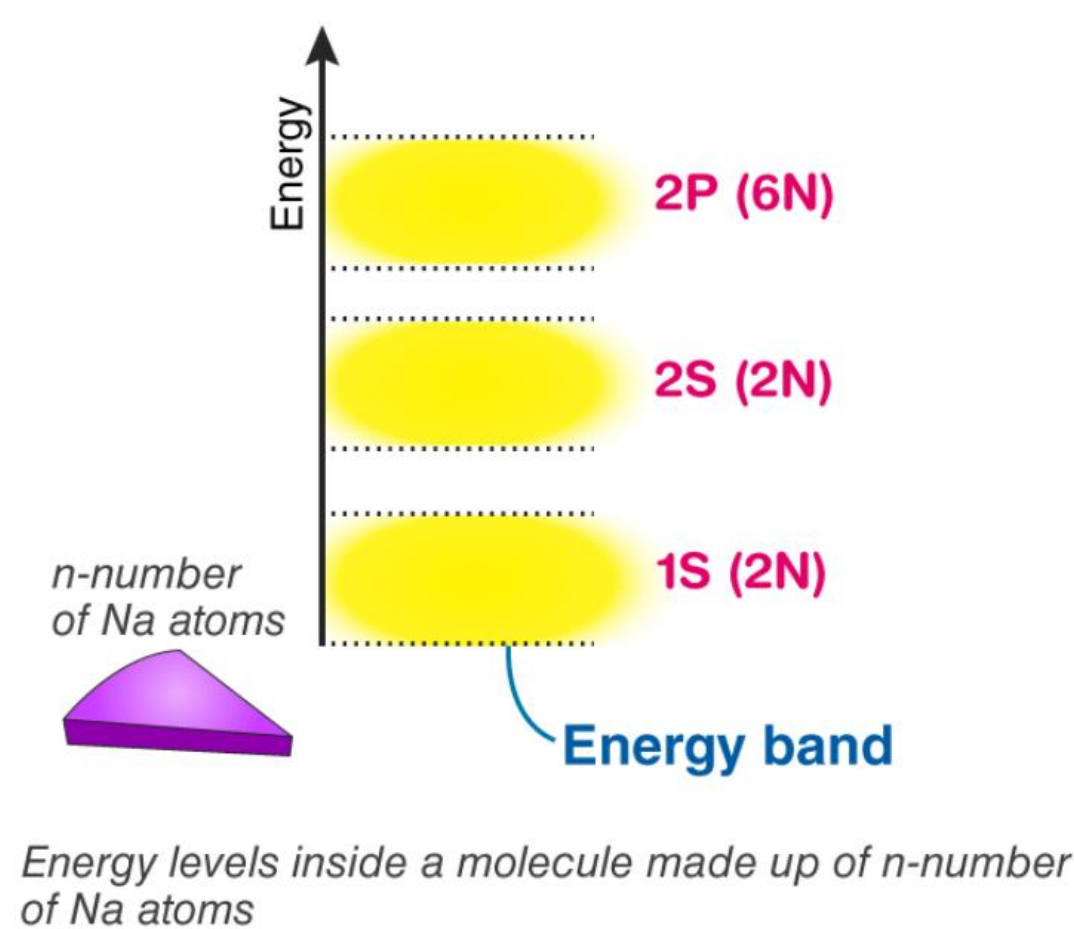


Energy levels inside a molecule made up of two Na atoms

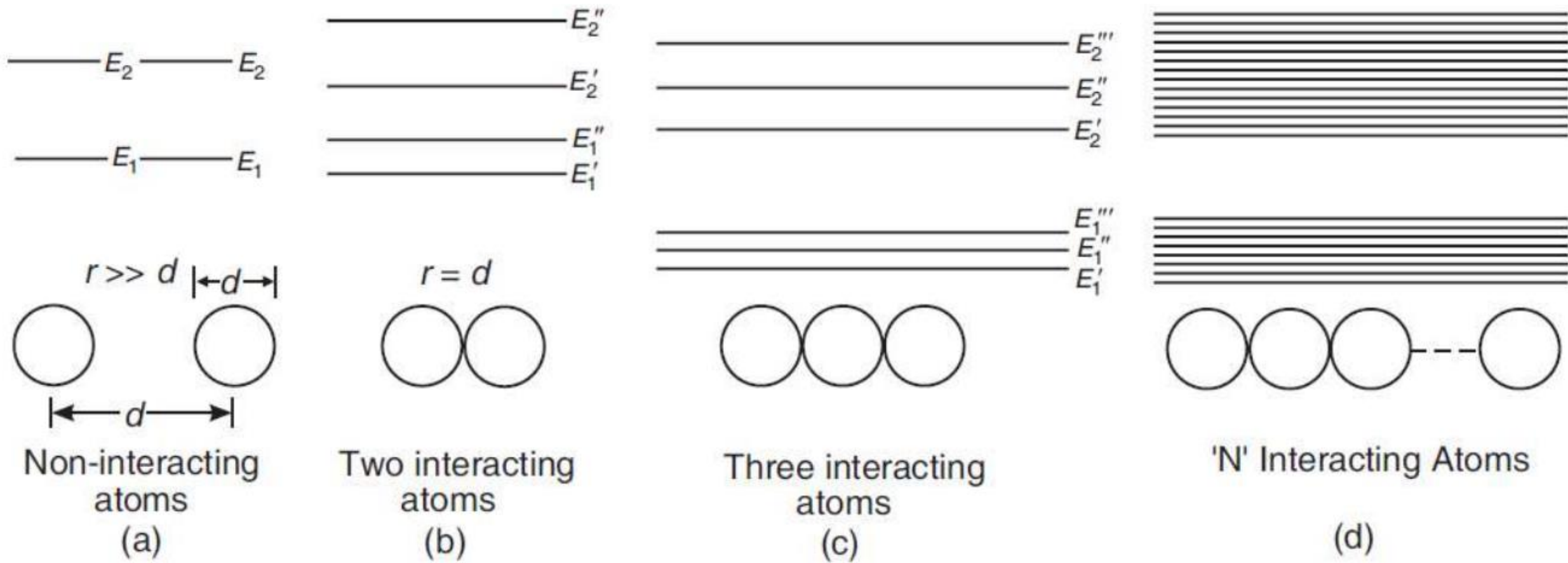


Energy levels inside a molecule made up of three Na atoms

Band Theory

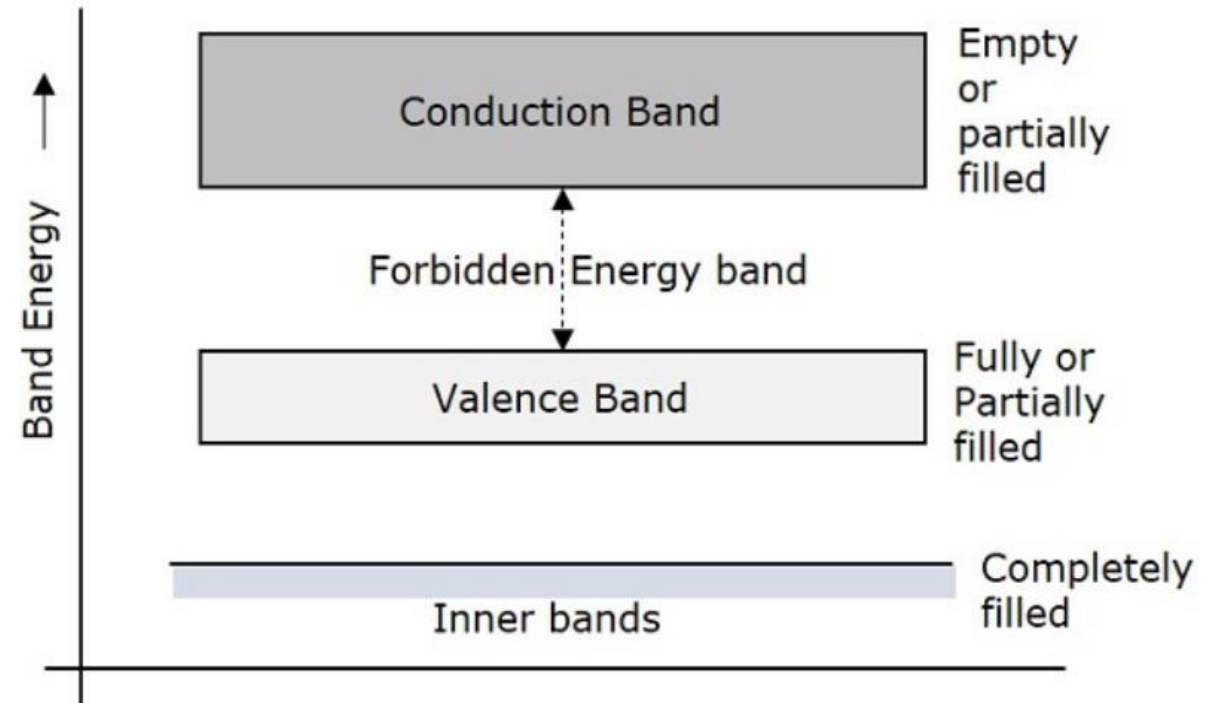
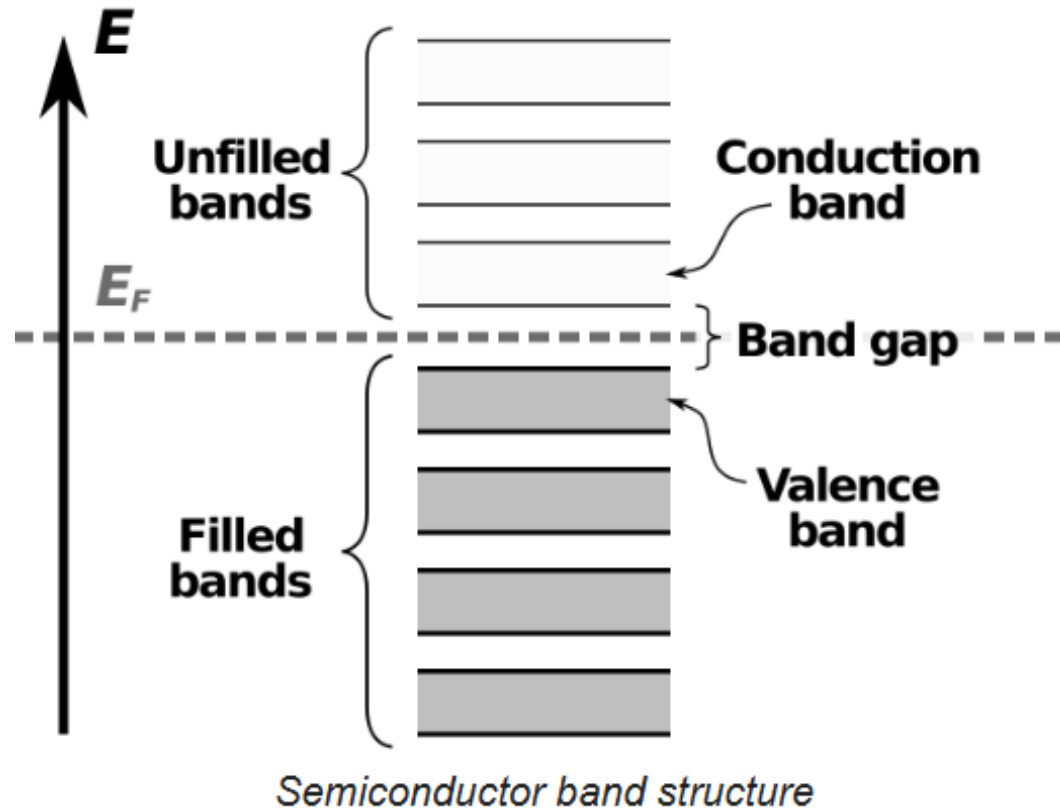


Band Theory

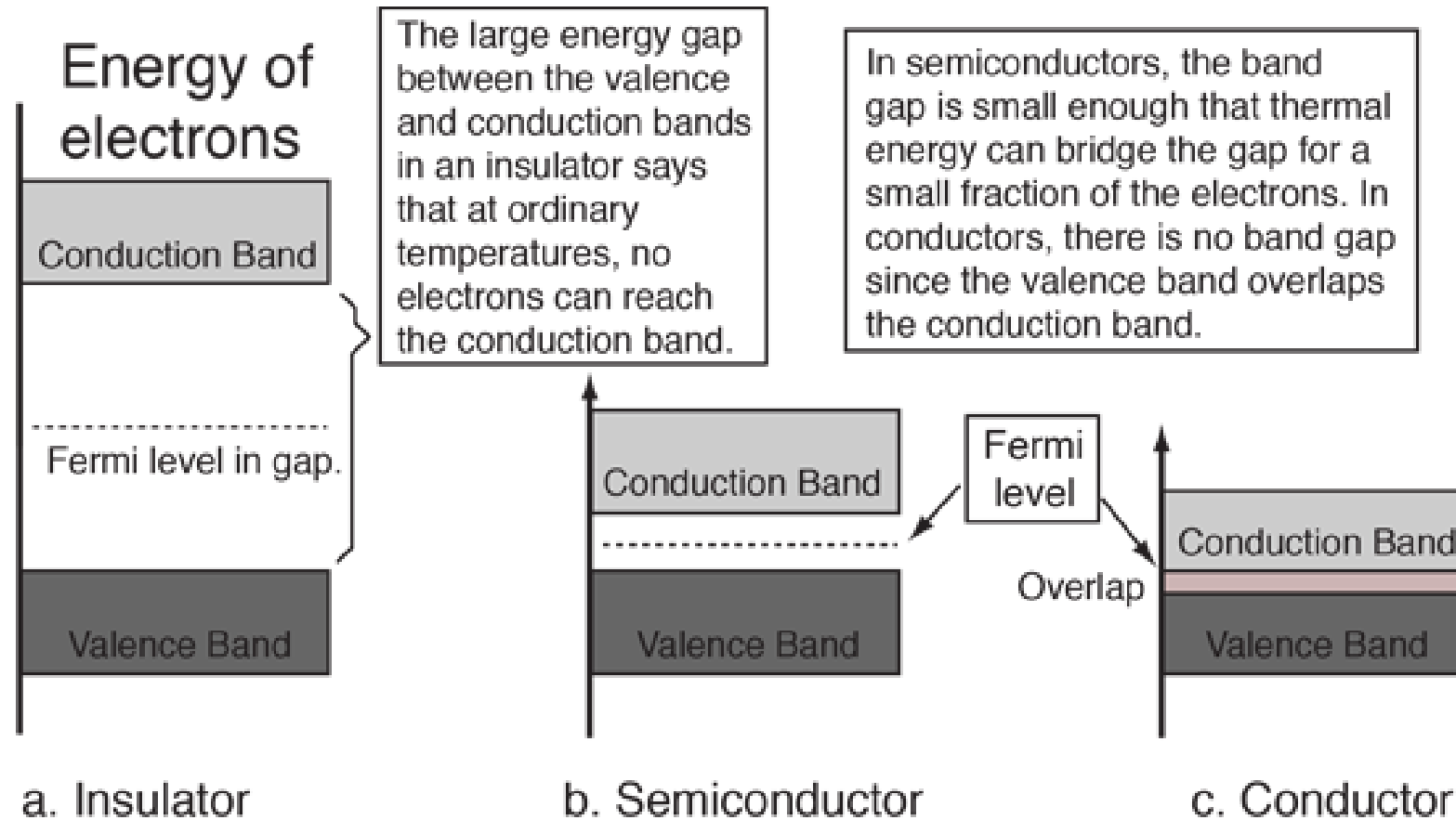


Band Theory

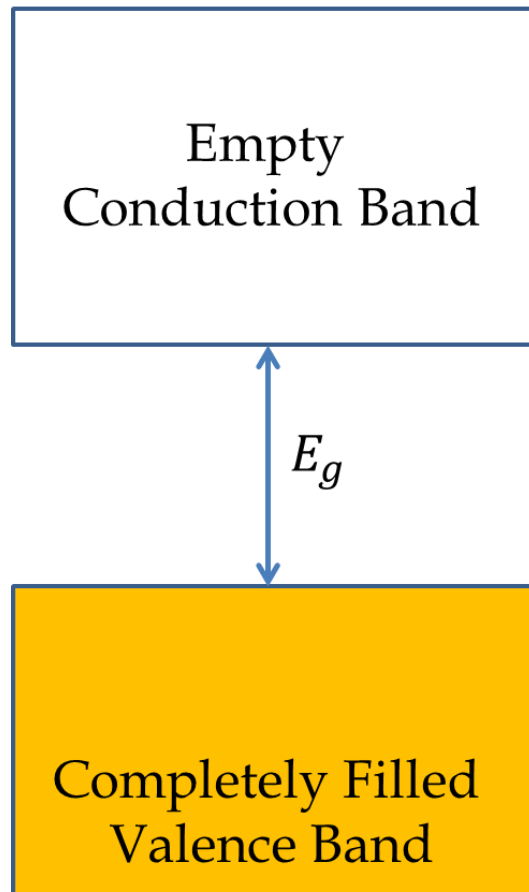
The **valence** band is the highest range of **electron energies** in which electrons are normally present at **absolute zero** temperature, while the conduction band is the lowest range of vacant **electronic states**.



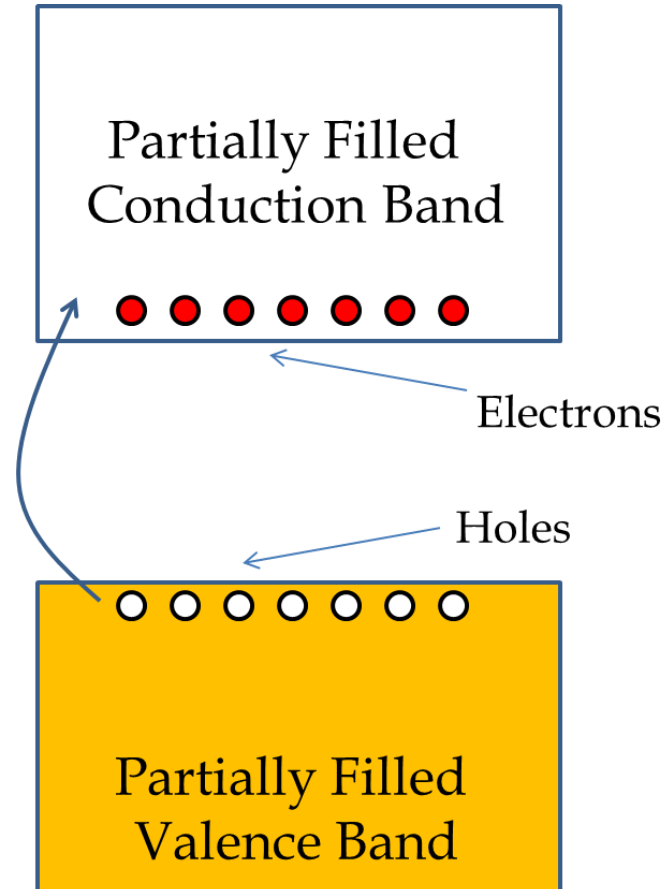
Classification of Materials



Understanding Conduction

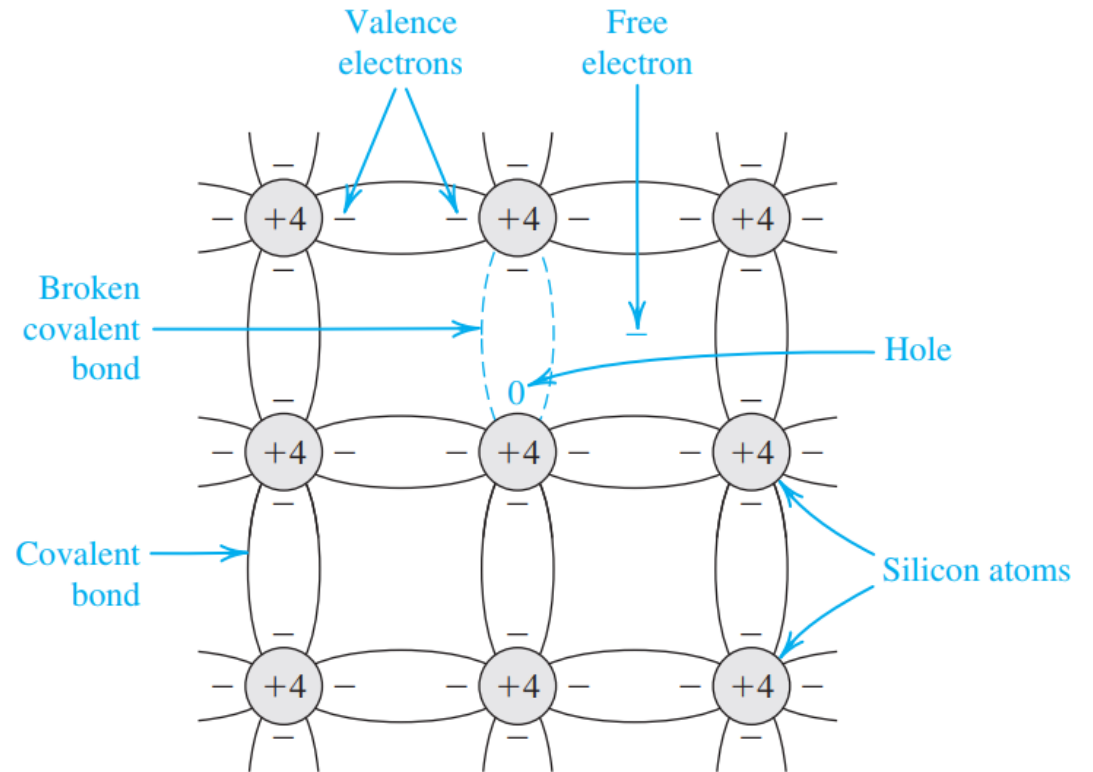
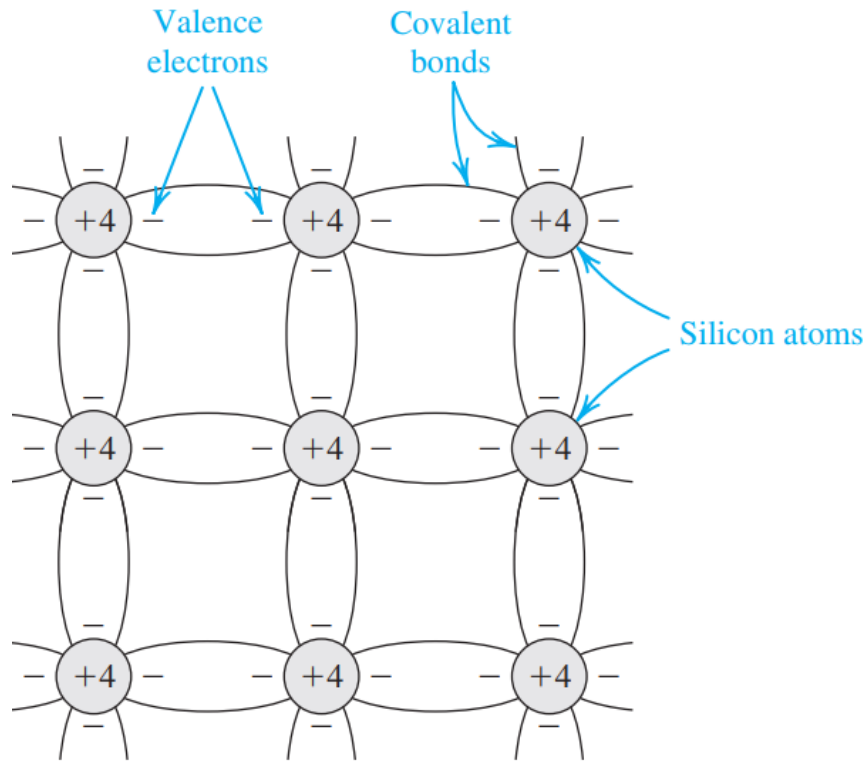


(a) $T = 0\text{ K}$



(a) $T > 0\text{ K}$

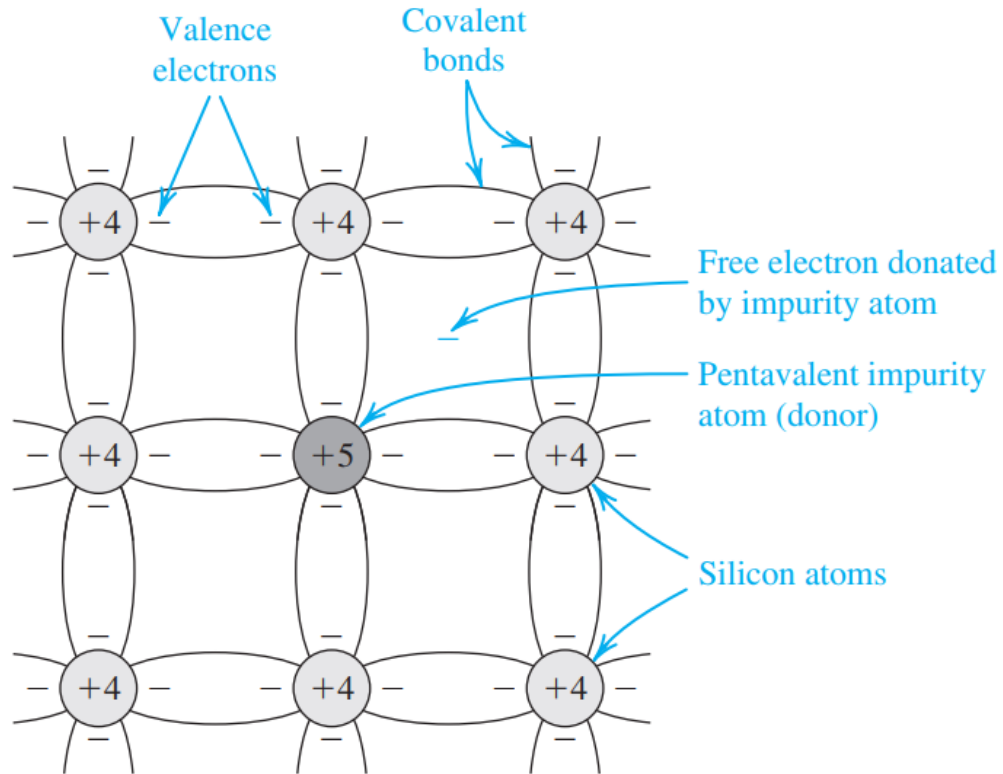
Intrinsic Semiconductor



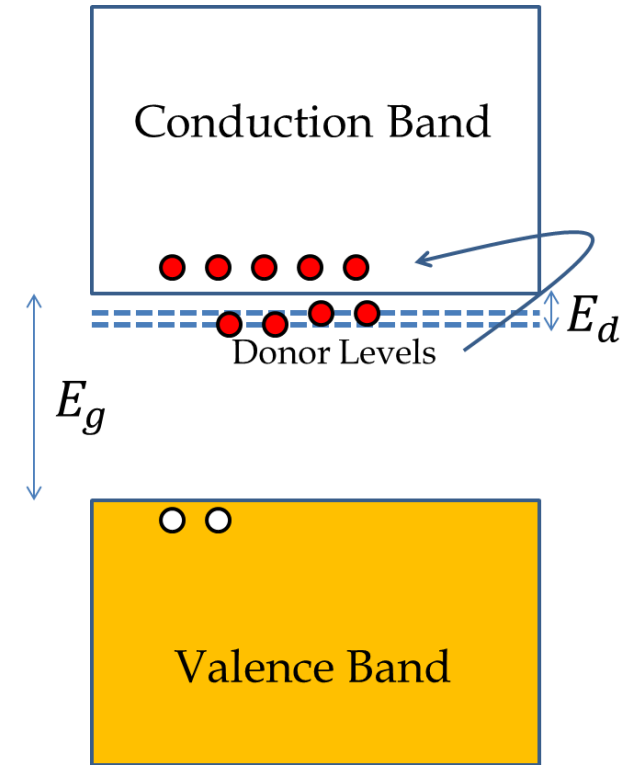
At room temperature, some of the covalent bonds are broken by thermal generation. Each broken bond gives rise to a free electron and a hole, both of which become available for current conduction.

$$n = p = n_i$$

Extrinsic Semiconductor: n-type



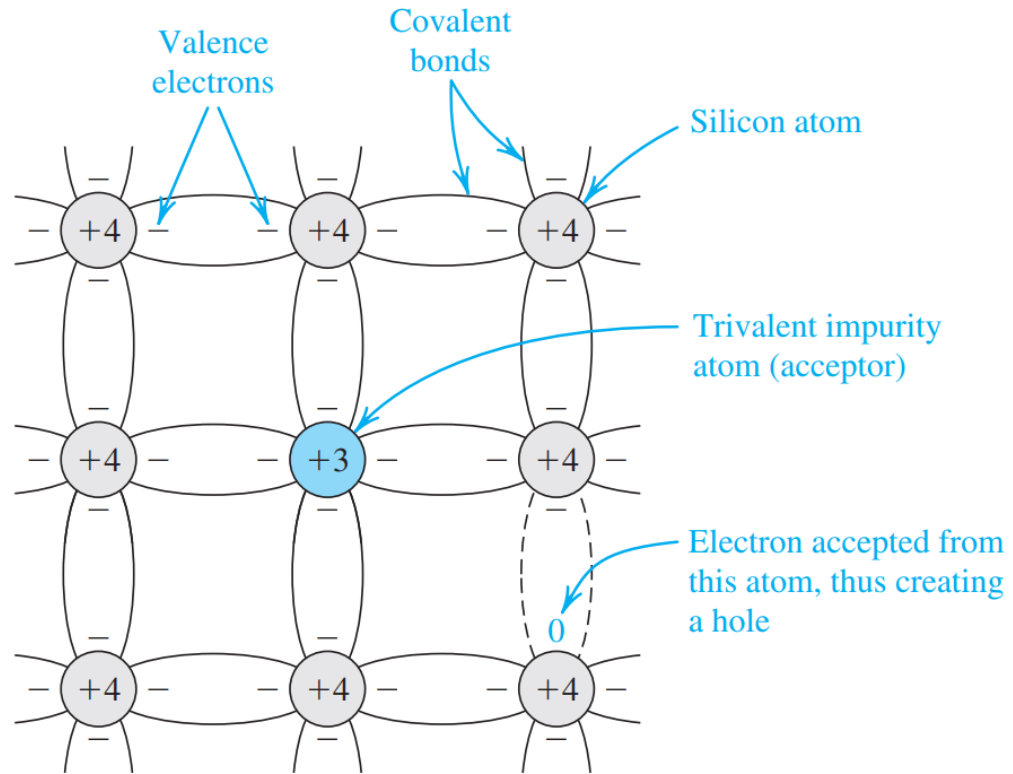
Majority Carrier: electron



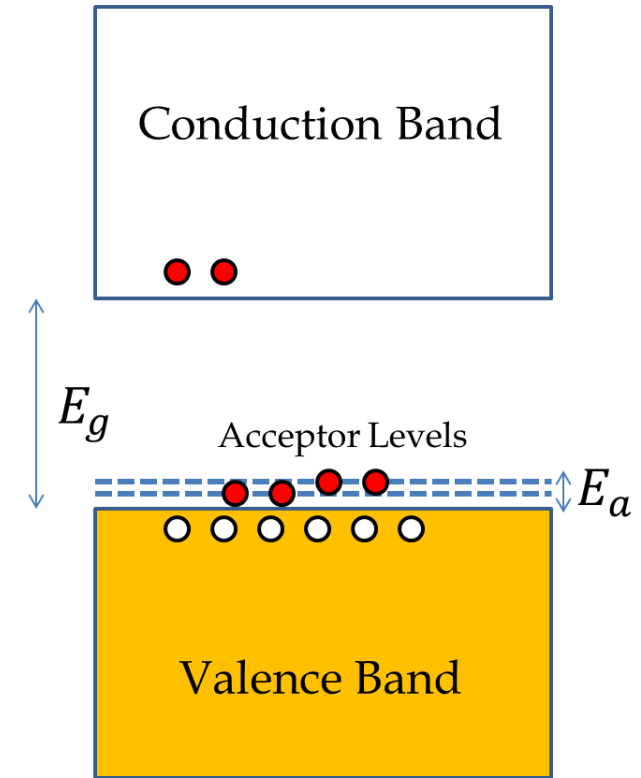
$$n \gg p$$

$$n_n \simeq N_D$$

Extrinsic Semiconductor: p-type



Majority Carrier: hole



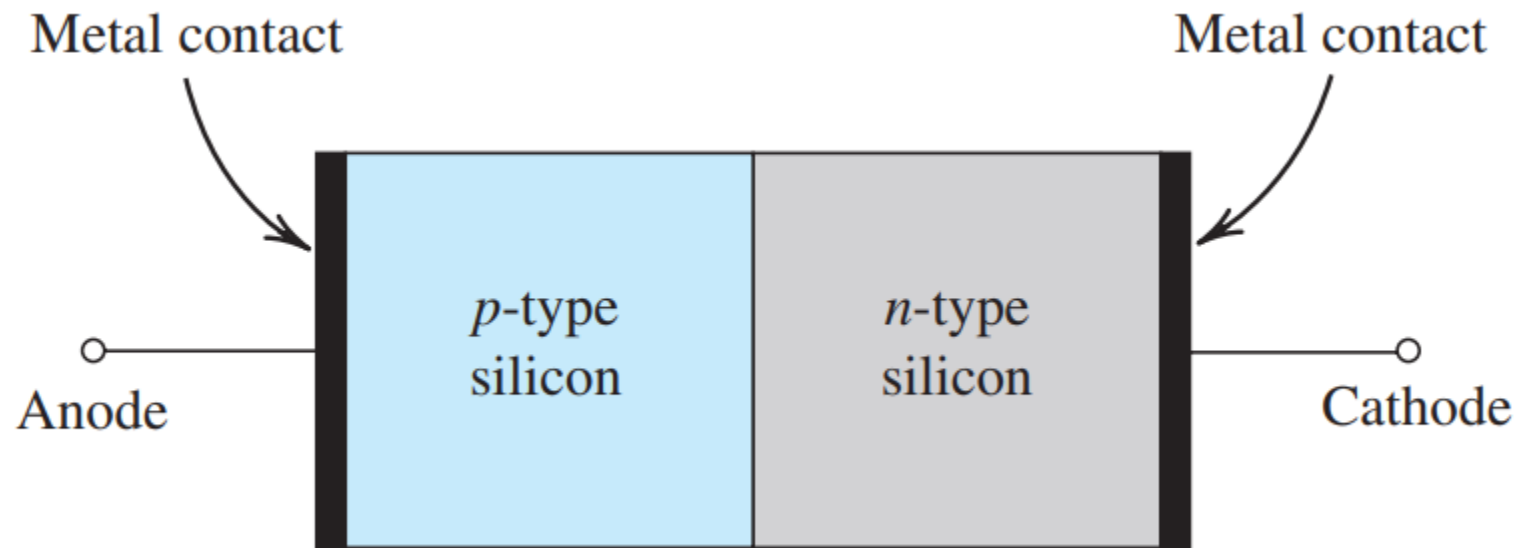
$$p \gg n$$

$$p_p \simeq N_A$$

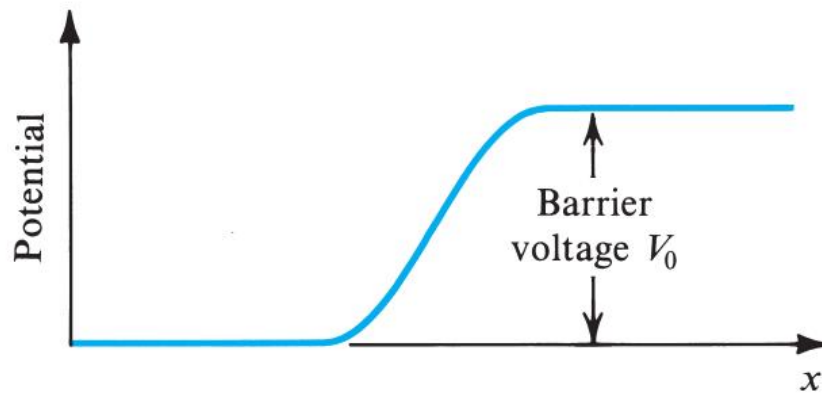
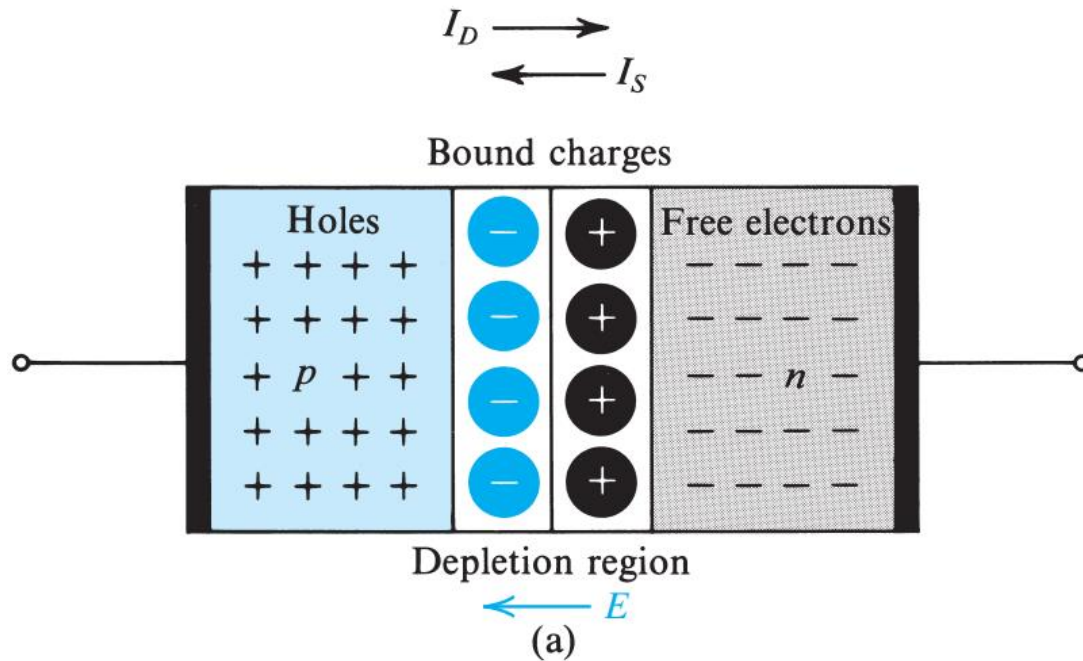
The pn Junction

Drift Current: Drift current can be defined as the charge carrier's moves in a semiconductor because of the electric field.

Diffusion Current: The diffusion current can be defined as the flow of charge carriers within a semiconductor travels from a higher concentration region to a lower concentration region.

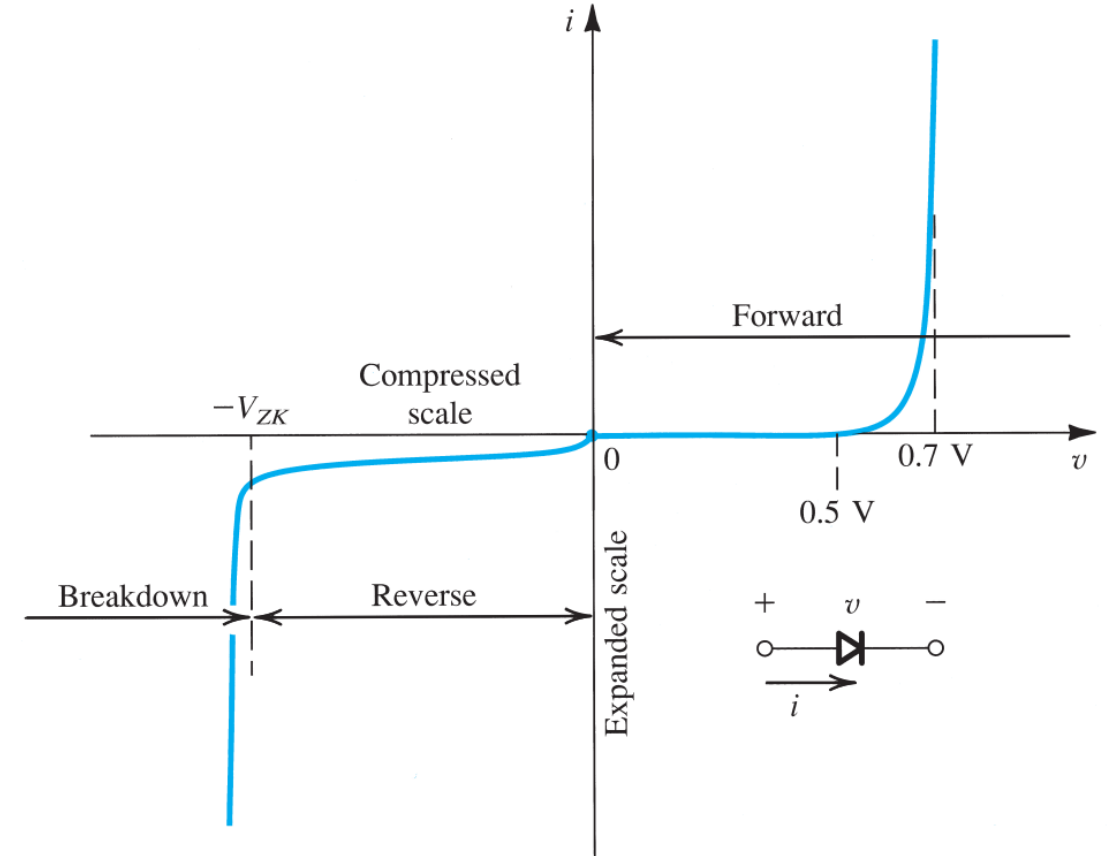
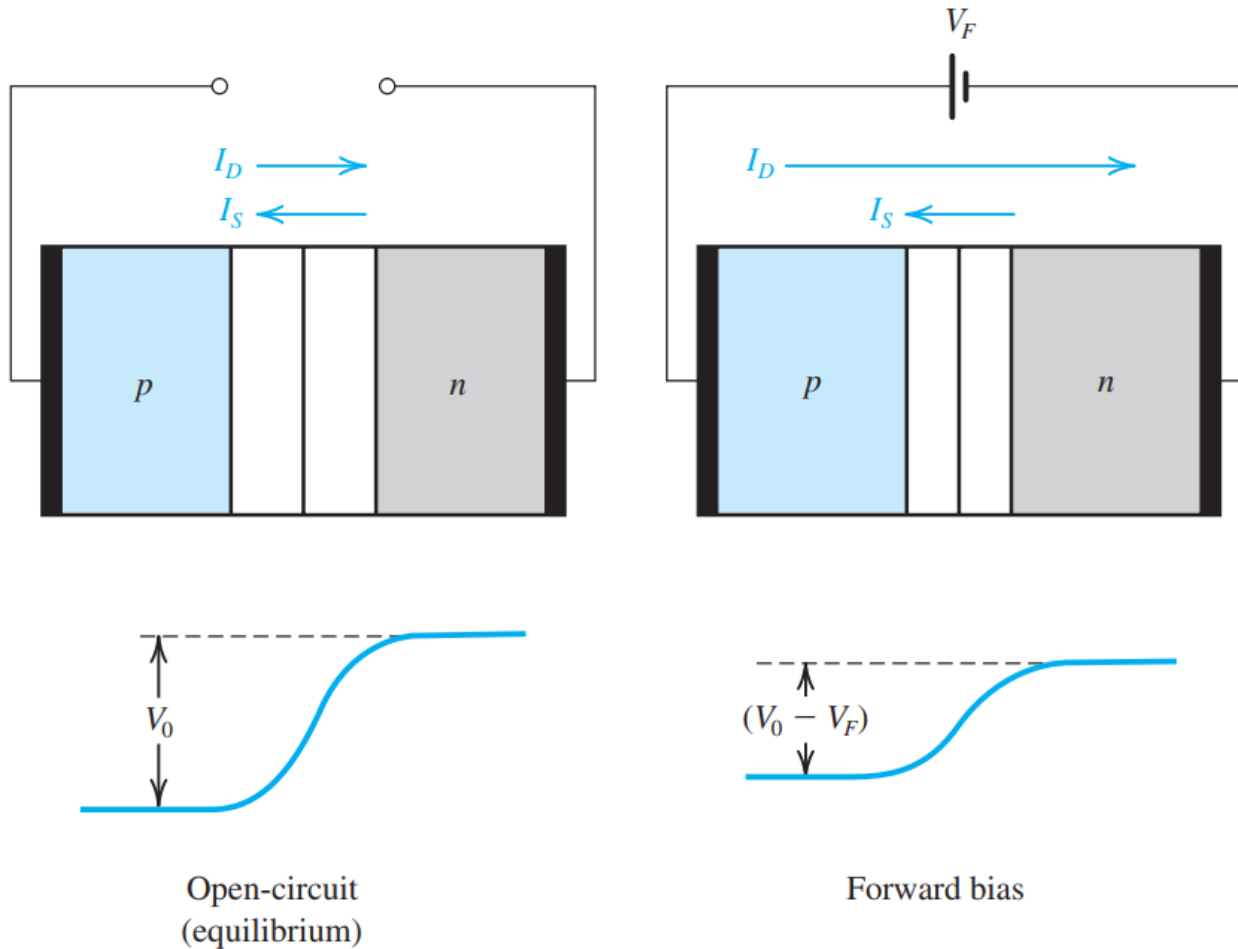


The pn Junction: No Applied Bias



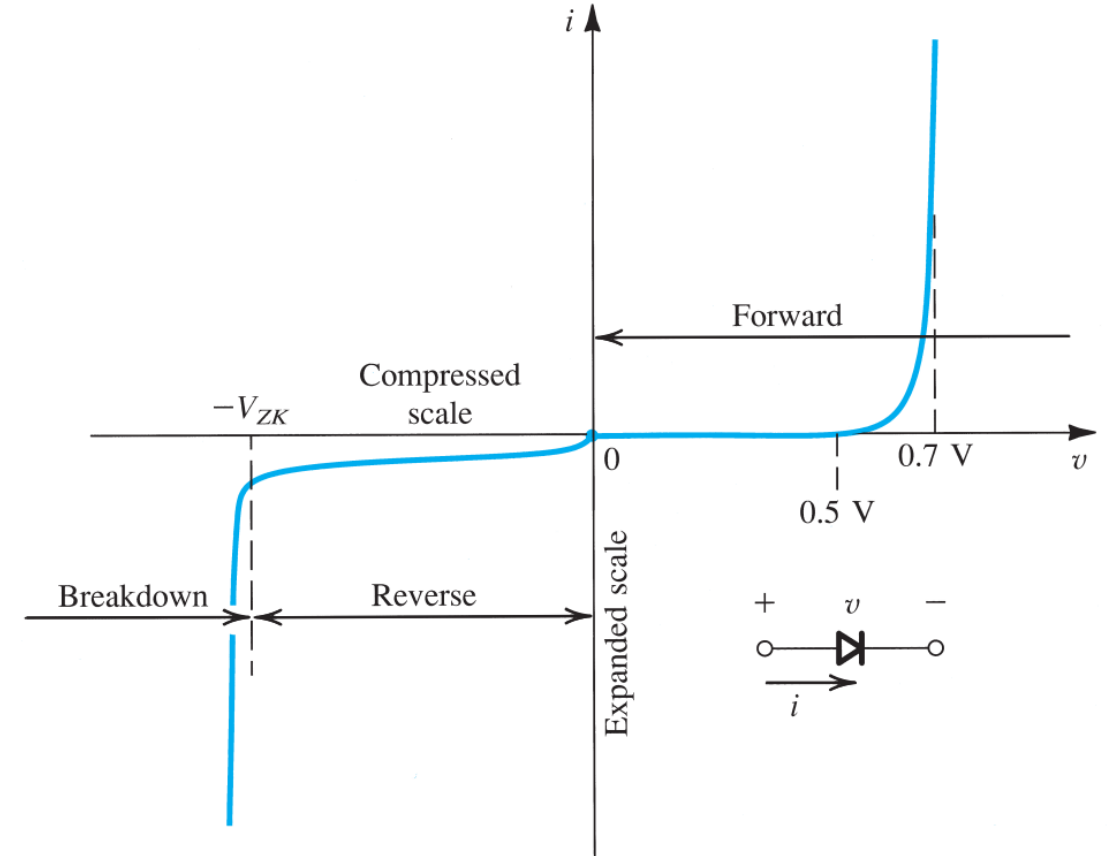
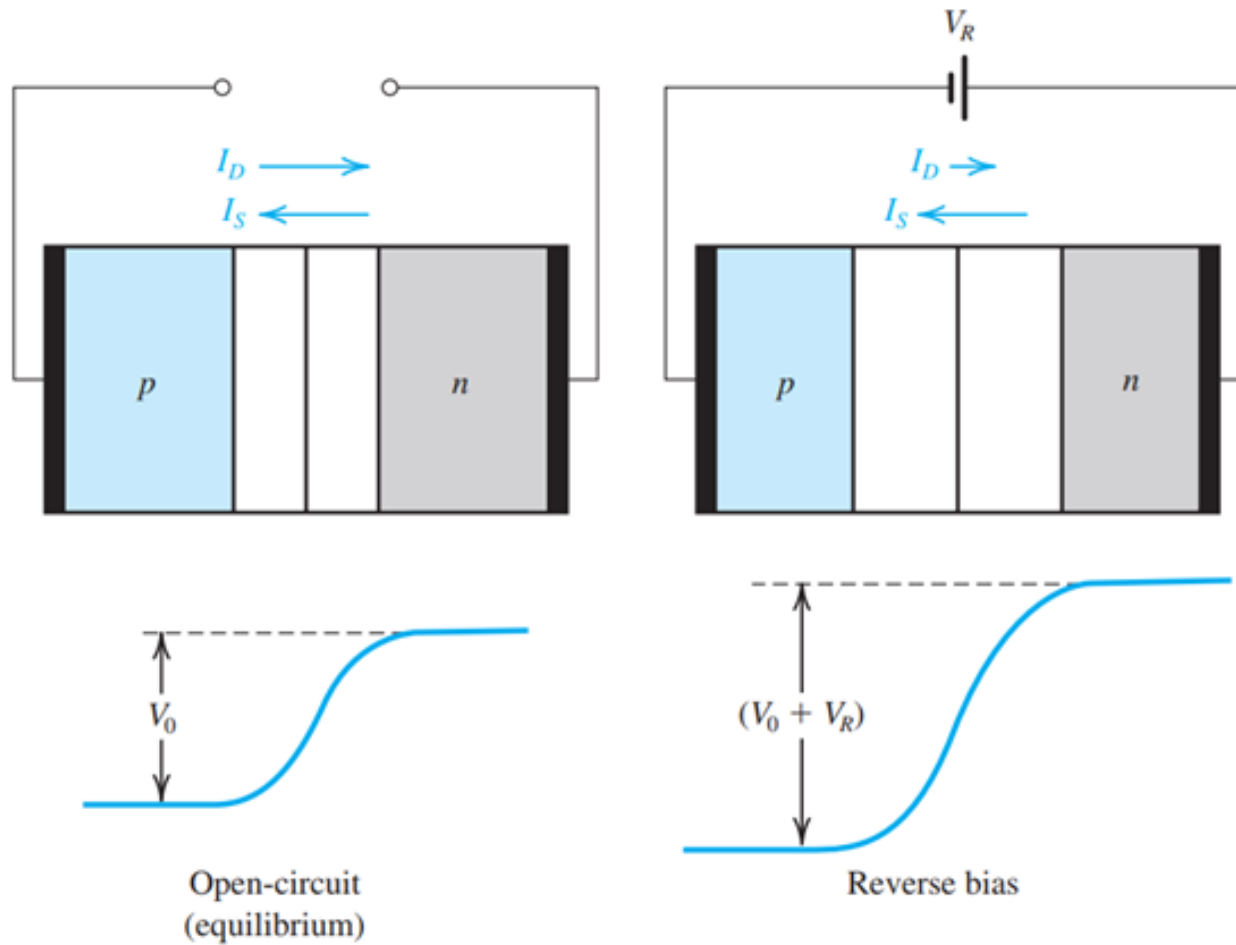
$$V_0 = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

The pn Junction: Forward Bias



$$i = I_S(e^{v/nV_T} - 1)$$

The pn Junction: Reverse Bias

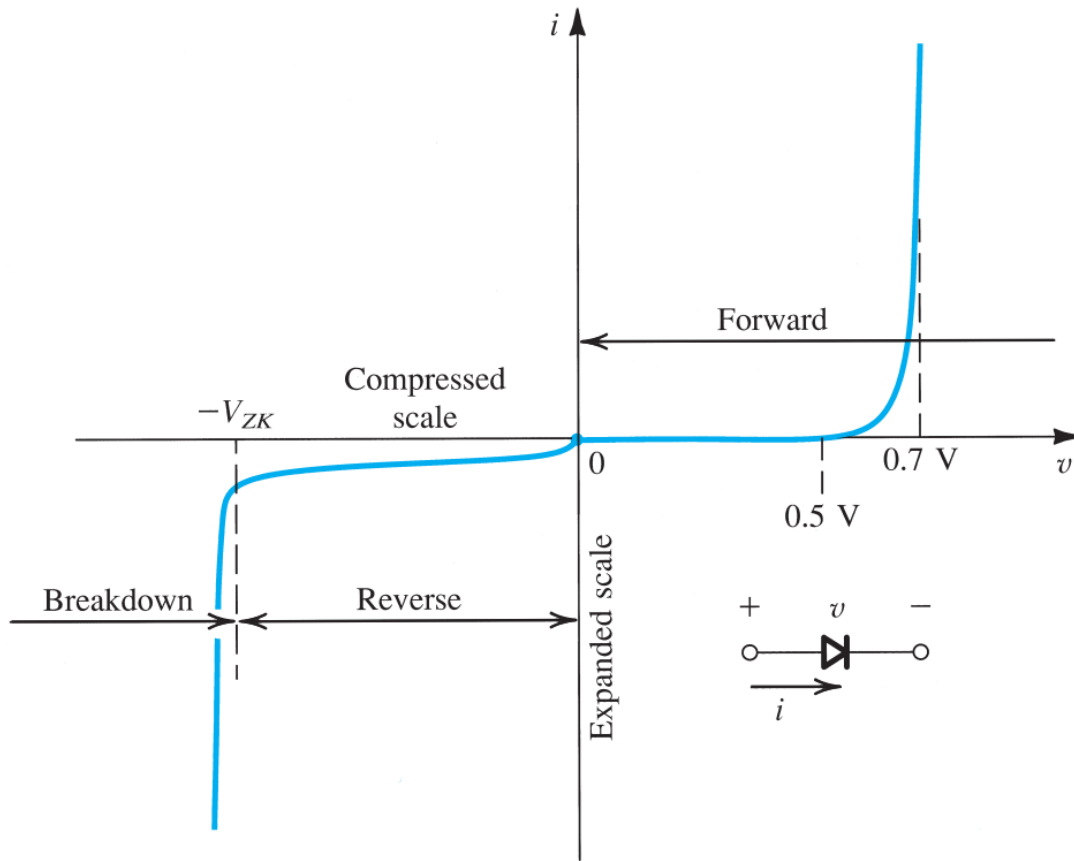


There are two possible mechanisms for pn junction breakdown:

1. Zener effect
2. Avalanche effect

$$i = I_S(e^{v/nV_T} - 1)$$

The pn Junction: I-V Characteristics



1. The forward-bias region, determined by $v > 0$
2. The reverse-bias region, determined by $v < 0$
3. The breakdown region, determined by $v < -V_{ZK}$

$$i = I_S(e^{v/nV_T} - 1)$$

with n having a value between 1 and 2, depending on the material and the physical structure of the diode.

$$V_T = \frac{kT}{q}$$

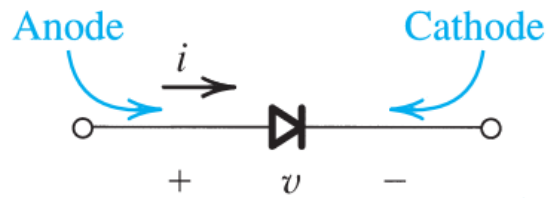
where

k = Boltzmann's constant = $8.62 \times 10^{-5} \text{ eV/K} = 1.38 \times 10^{-23} \text{ joules/kelvin}$

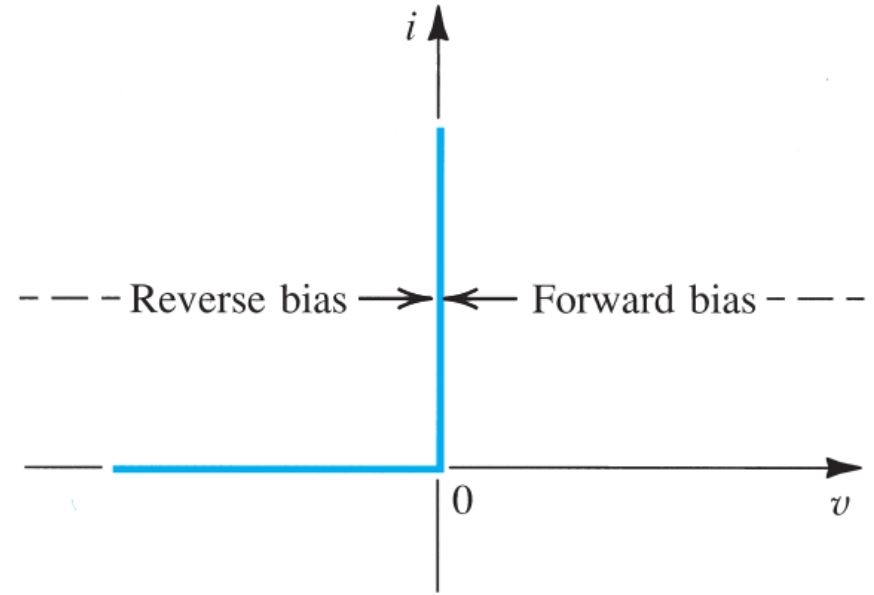
T = the absolute temperature in kelvins = $273 + \text{temperature in } ^\circ\text{C}$

q = the magnitude of electronic charge = $1.60 \times 10^{-19} \text{ coulomb}$

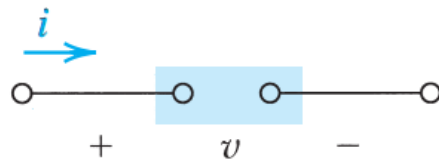
Ideal Diode Model



(a)

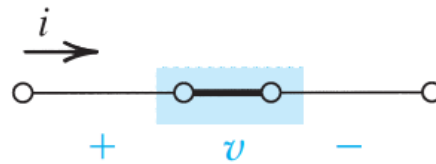


(b)



$$v < 0 \Rightarrow i = 0$$

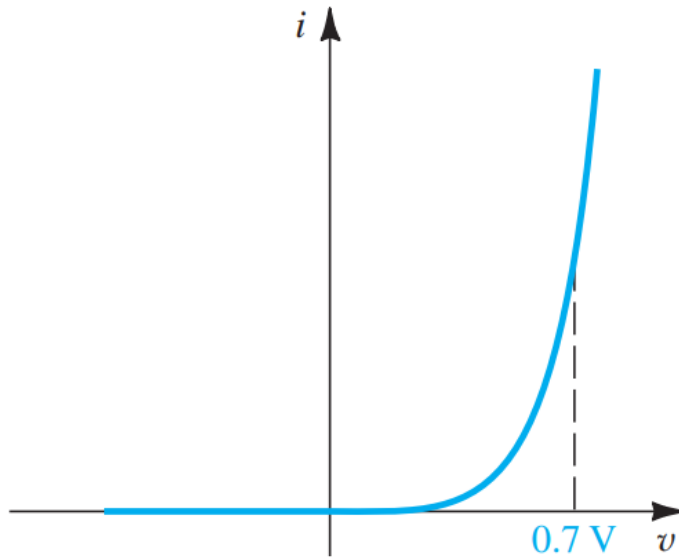
(c)



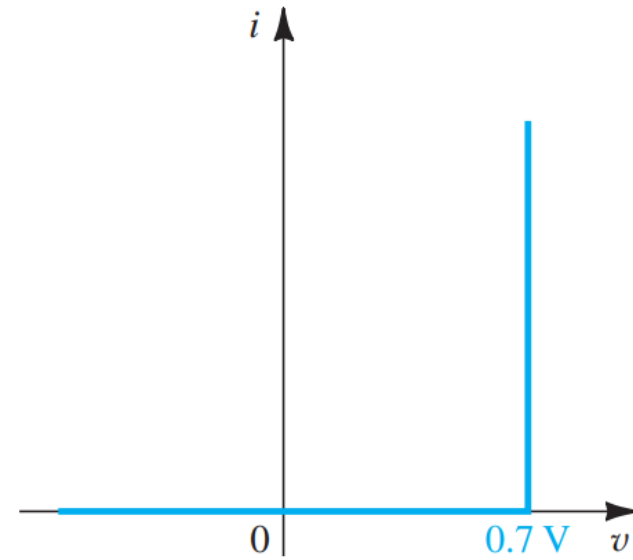
$$i > 0 \Rightarrow v = 0$$

(d)

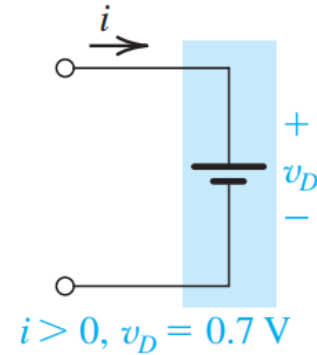
Constant Voltage-Drop Model



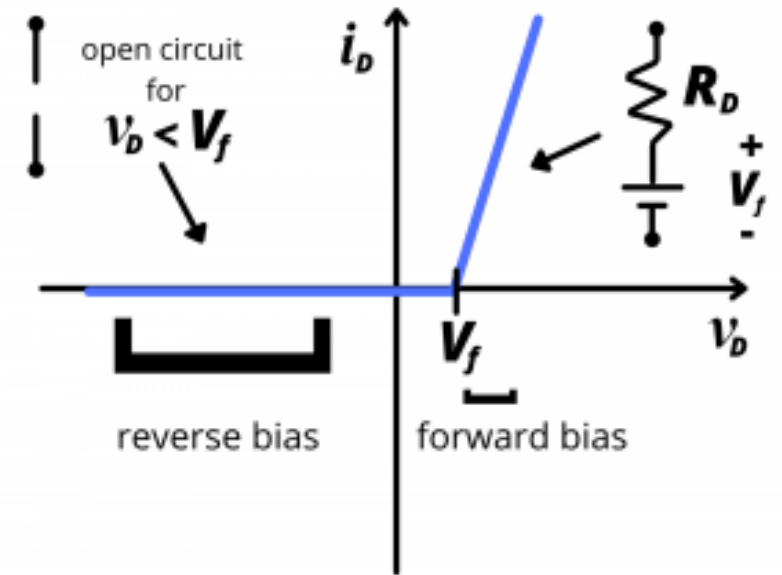
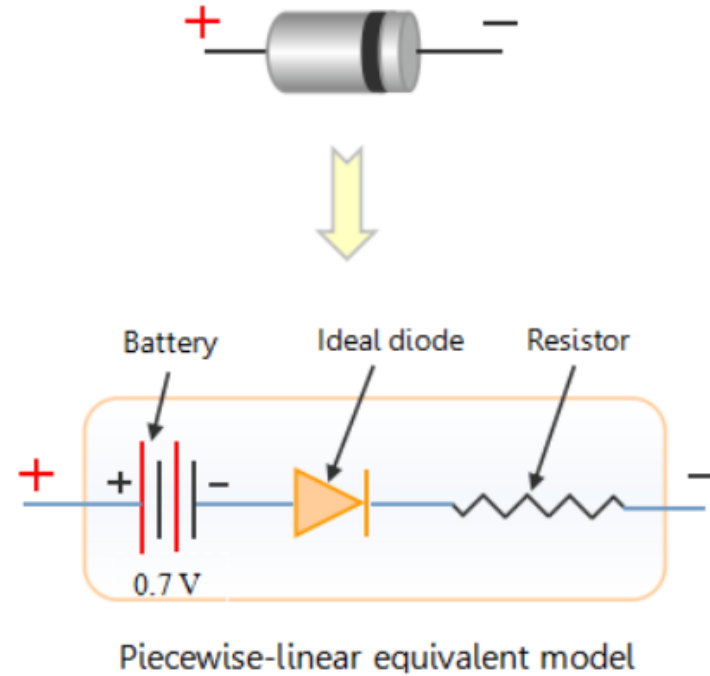
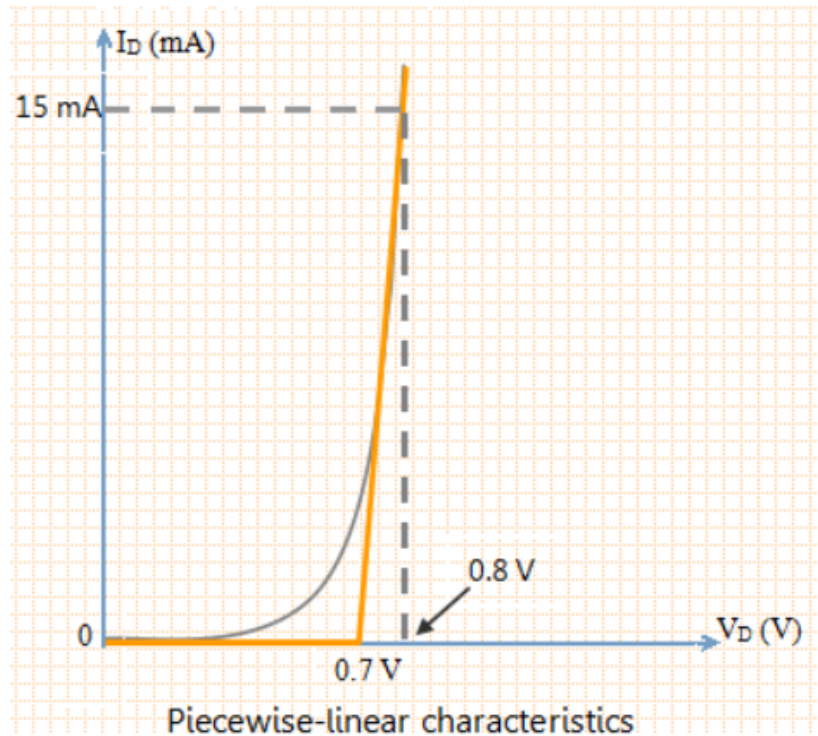
(a)



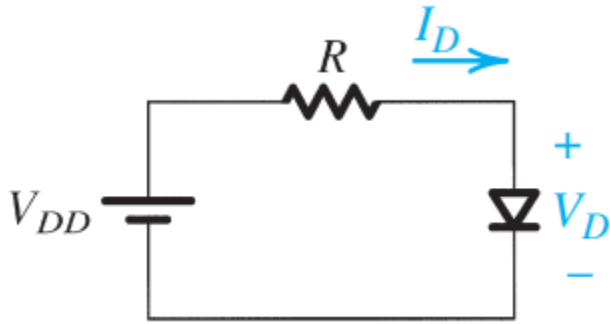
(b)



Piecewise Linear Model



Exponential Model

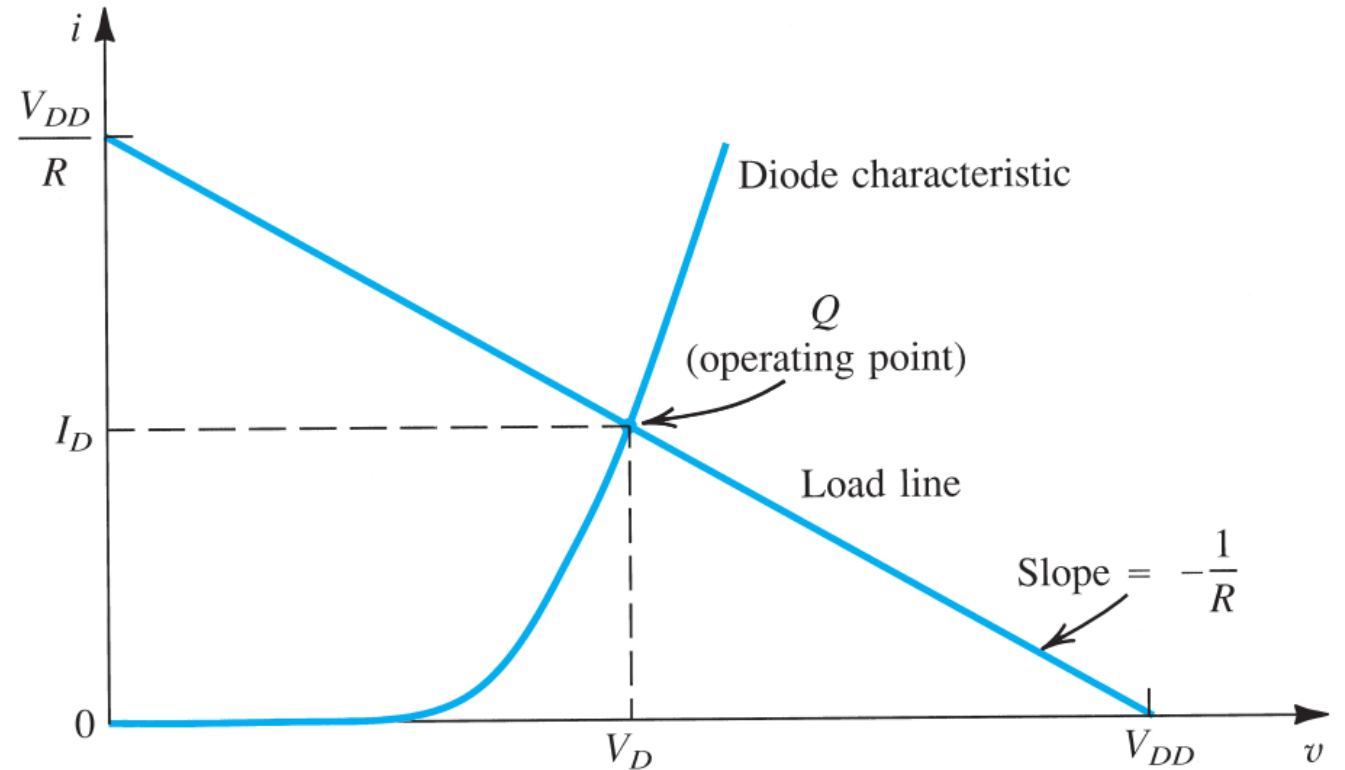


$$I_D = I_S e^{V_D/V_T}$$

$$I_D = \frac{V_{DD} - V_D}{R}$$

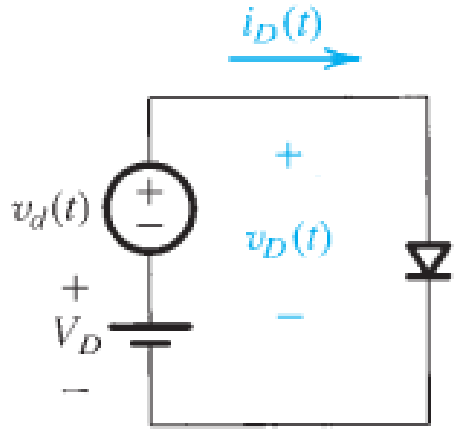
Two alternative ways for obtaining the solution are:

1. **Graphical Analysis**
2. **Iterative Analysis**



Graphical analysis of the circuit using the exponential diode model.

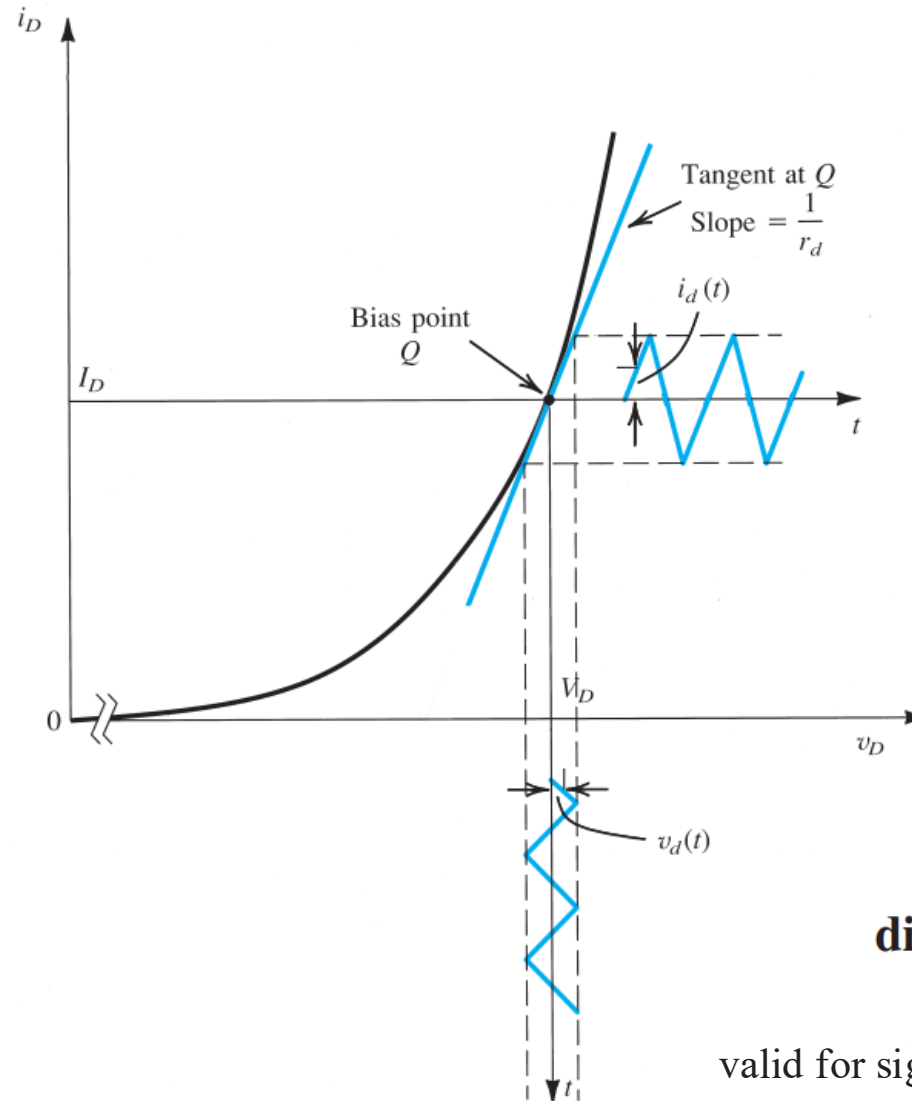
Small Signal Model



$$v_D(t) = V_D + v_d(t)$$

$$i_D = I_D + i_d$$

$$\frac{v_d}{V_T} \ll 1$$



$$i_D(t) = I_S e^{v_D/V_T}$$

$$i_D(t) = I_S e^{(V_D + v_d)/V_T}$$

$$i_D(t) = I_S e^{V_D/V_T} e^{v_d/V_T}$$

$$i_D(t) = I_D e^{v_d/V_T}$$

$$i_D(t) \simeq I_D \left(1 + \frac{v_d}{V_T} \right)$$

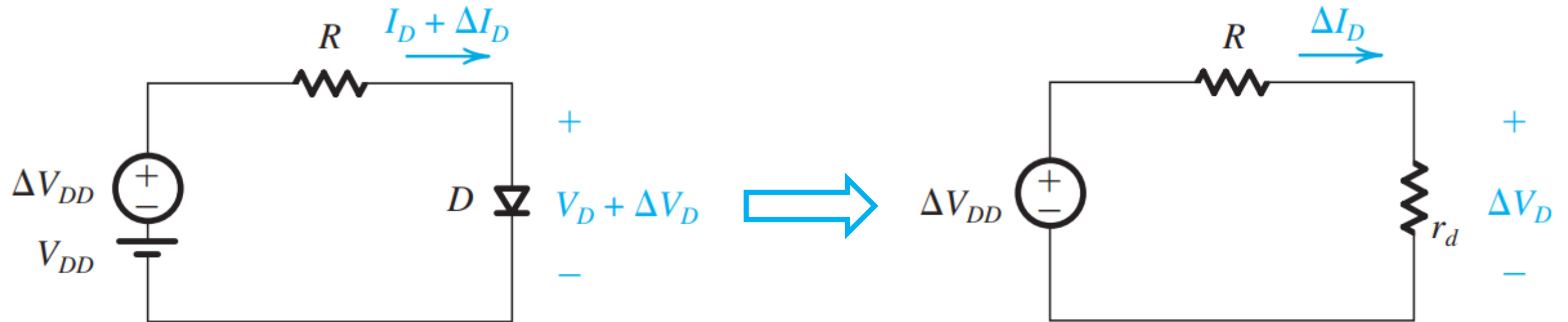
$$I_D + i_d = I_D + \frac{I_D}{V_T} v_d$$

$$i_d = \frac{I_D}{V_T} v_d$$

$$\text{diode small-signal resistance } r_d = \frac{V_T}{I_D}$$

valid for signals whose amplitudes are smaller than about 5 mV

Small Signal Model

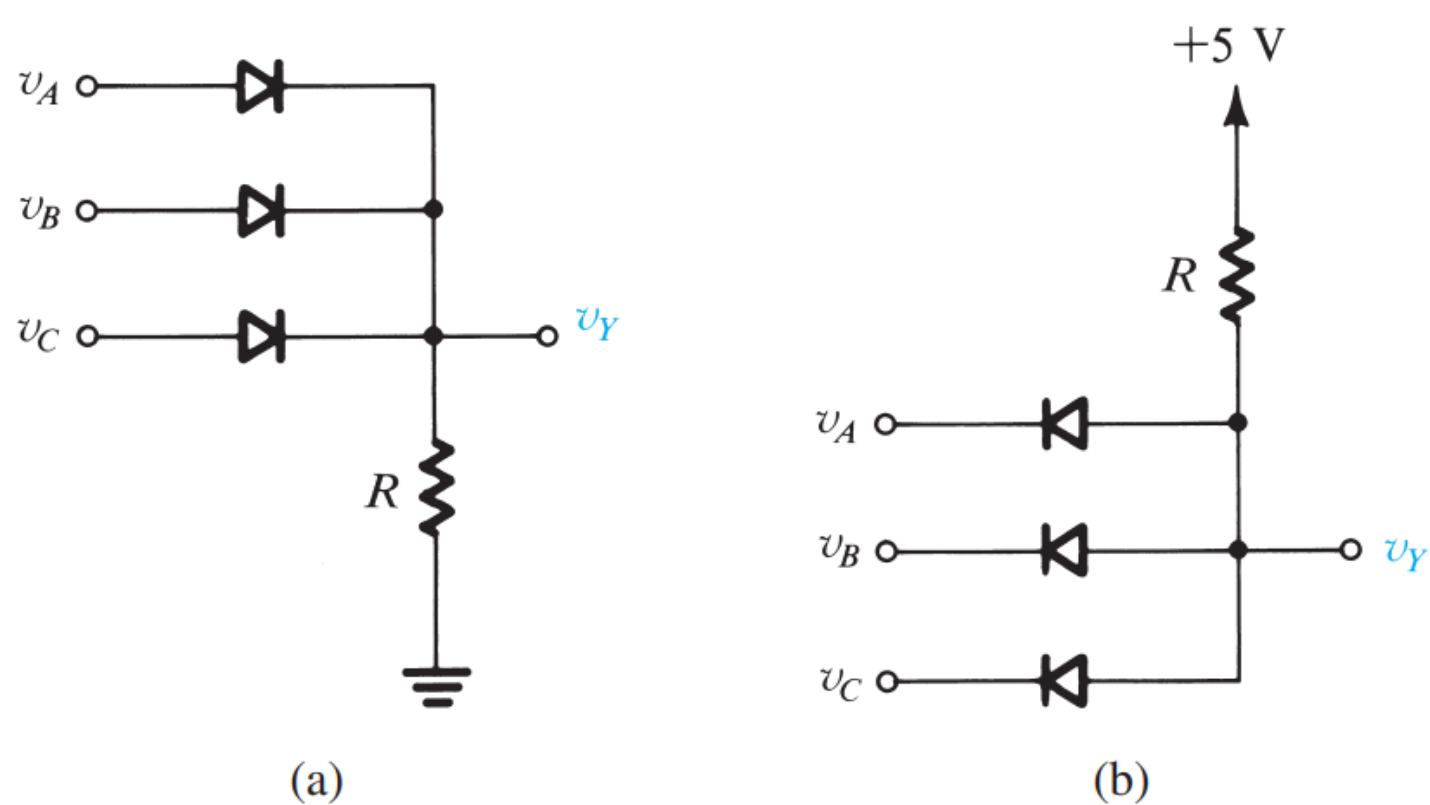


$$\frac{v_d}{V_T} \ll 1$$

$$r_d = 1 / \left[\frac{\partial i_D}{\partial v_D} \right]_{i_D = I_D}$$

$$r_d = \frac{V_T}{I_D}$$

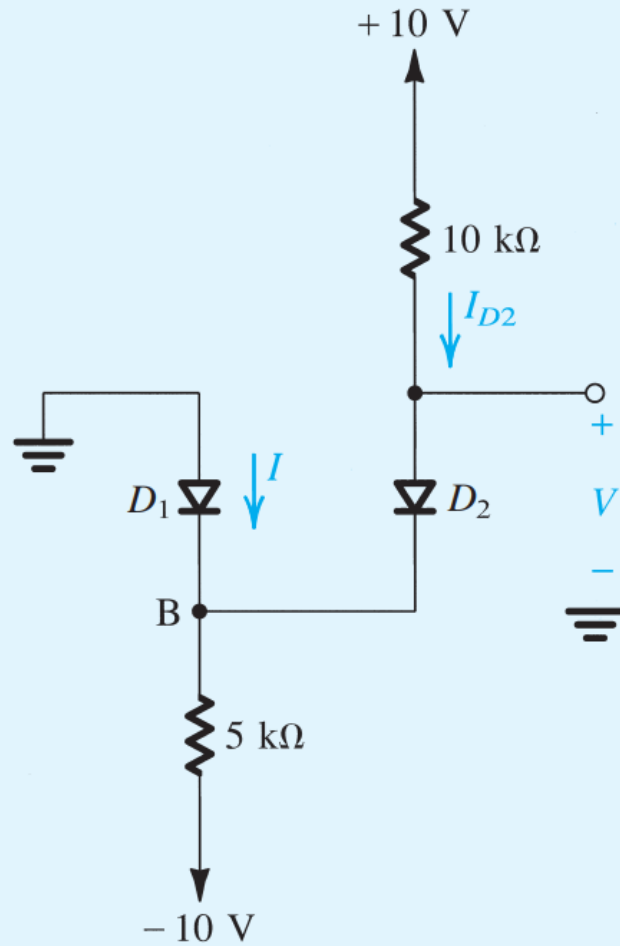
Diode Circuit Analysis



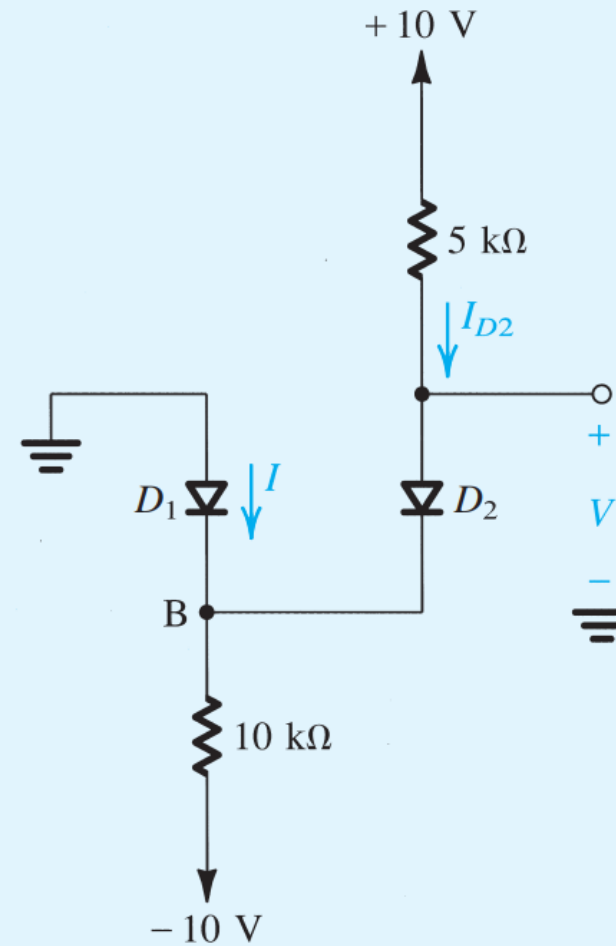
Diode logic gates: (a) OR gate; (b) AND gate (in a positive-logic system).

Diode Circuit Analysis

Assuming the diodes to be ideal, find the values of I and V in the following circuits.



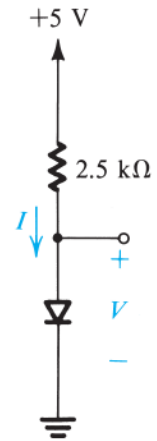
(a)



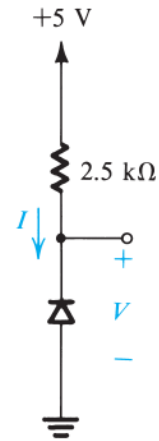
(b)

Diode Circuit Analysis

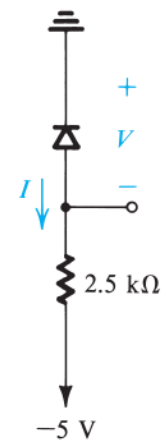
Find the values of I and V in the circuits shown in Figures below.



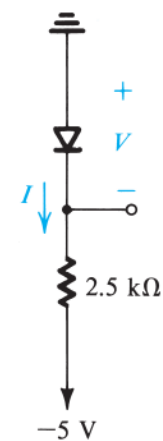
(a)



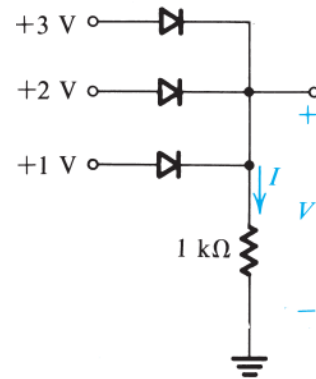
(b)



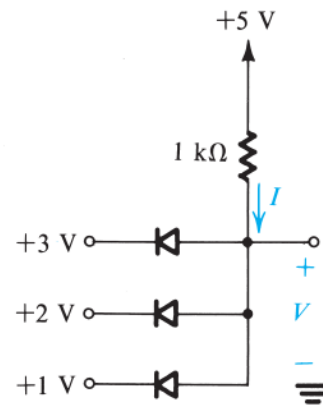
(c)



(d)



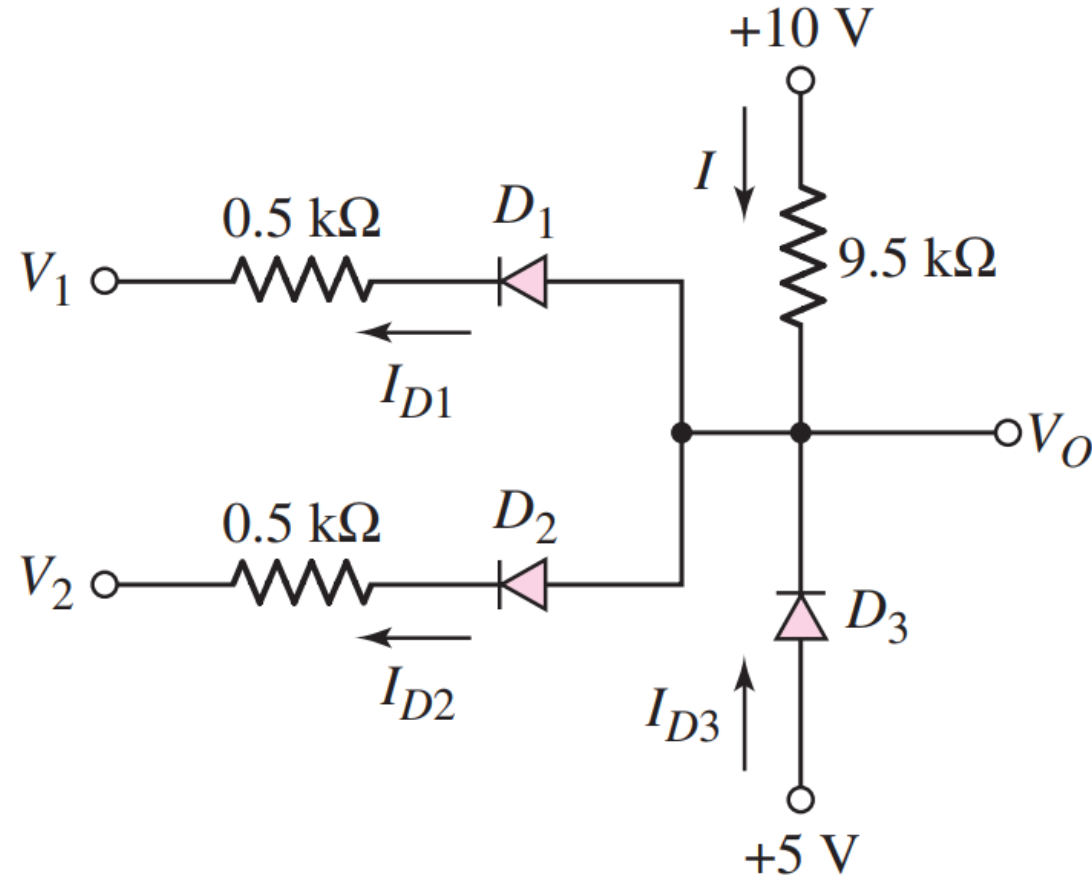
(e)



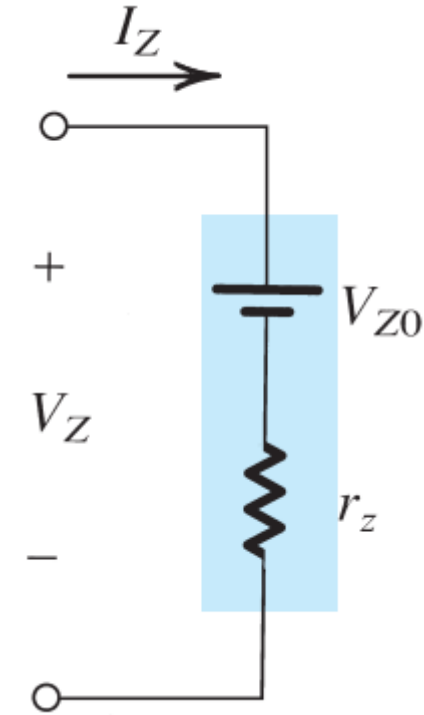
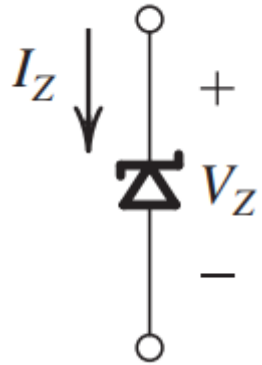
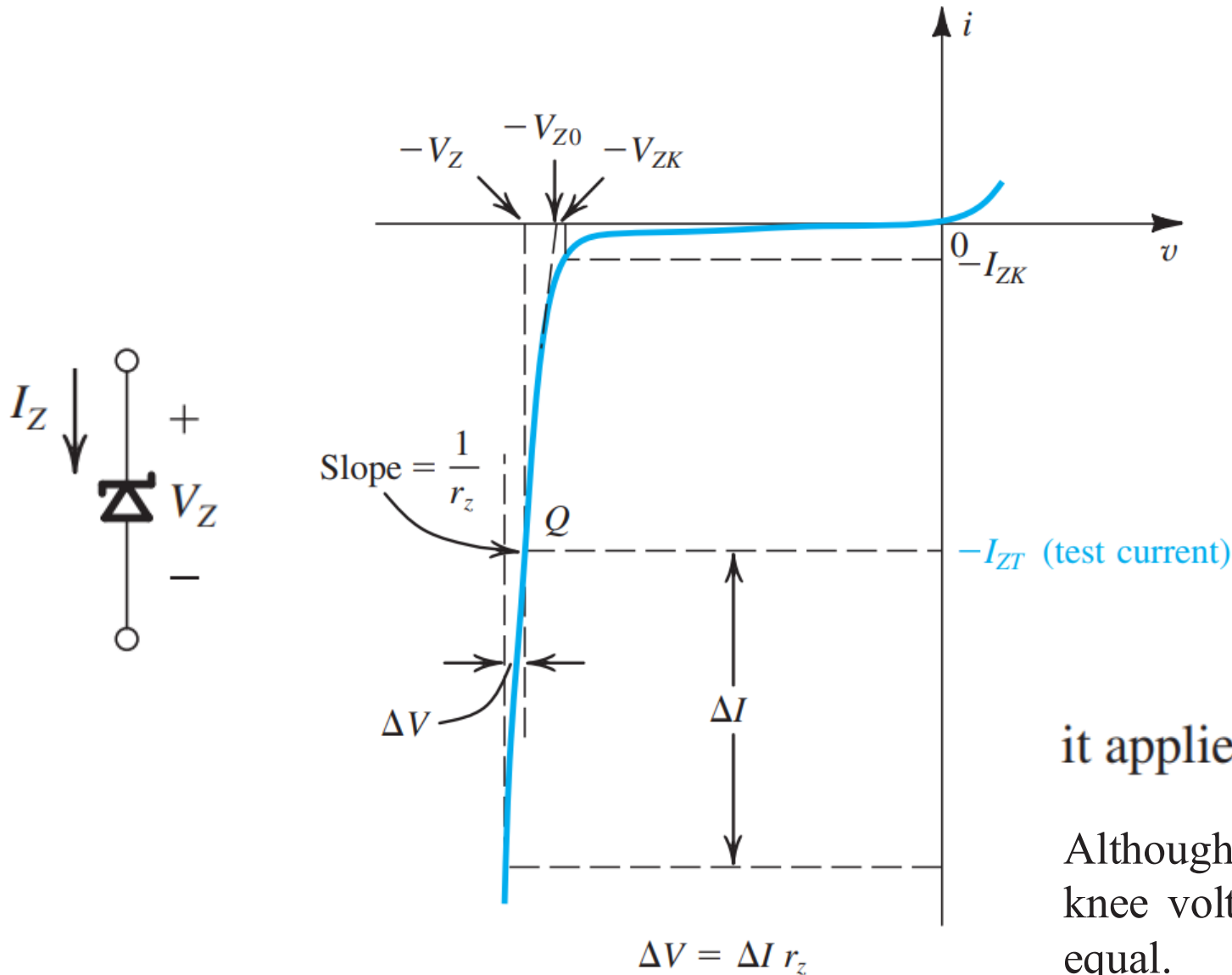
(f)

Diode Circuit Analysis

Determine the output voltage V_o and the currents I_{D1} , I_{D2} , I_{D3} , and I for the following input conditions: (a) $V_1 = V_2 = 0$; (b) $V_1 = V_2 = 5$ V; (c) $V_1 = 5$ V, $V_2 = 0$; and (d) $V_1 = 5$ V, $V_2 = 2$ V. The diodes have a constant voltage drop of 0.6V



Zener Diode



$$V_Z = V_{Z0} + r_z I_Z$$

it applies for $I_Z > I_{ZK}$ and, obviously, $V_Z > V_{Z0}$

Although V_{Z0} is shown to be slightly different from the knee voltage V_{ZK} , in practice their values are almost equal.

Zener Diode

The 6.8-V zener diode in the circuit of Fig. 4.21(a) is specified to have $V_Z = 6.8\text{ V}$ at $I_Z = 5\text{ mA}$, $r_z = 20\ \Omega$, and $I_{ZK} = 0.2\text{ mA}$. The supply voltage V^+ is nominally 10 V but can vary by $\pm 1\text{ V}$.

- (a) Find V_O with no load and with V^+ at its nominal value.
- (b) Find the change in V_O resulting from the $\pm 1\text{-V}$ change in V^+ . Note that $(\Delta V_O / \Delta V^+)$, usually expressed in mV/V, is known as **line regulation**.
- (c) Find the change in V_O resulting from connecting a load resistance R_L that draws a current $I_L = 1\text{ mA}$, and hence find the **load regulation** $(\Delta V_O / \Delta I_L)$ in mV/mA.
- (d) Find the change in V_O when $R_L = 2\text{ k}\Omega$.
- (e) Find the value of V_O when $R_L = 0.5\text{ k}\Omega$.
- (f) What is the minimum value of R_L for which the diode still operates in the breakdown region?

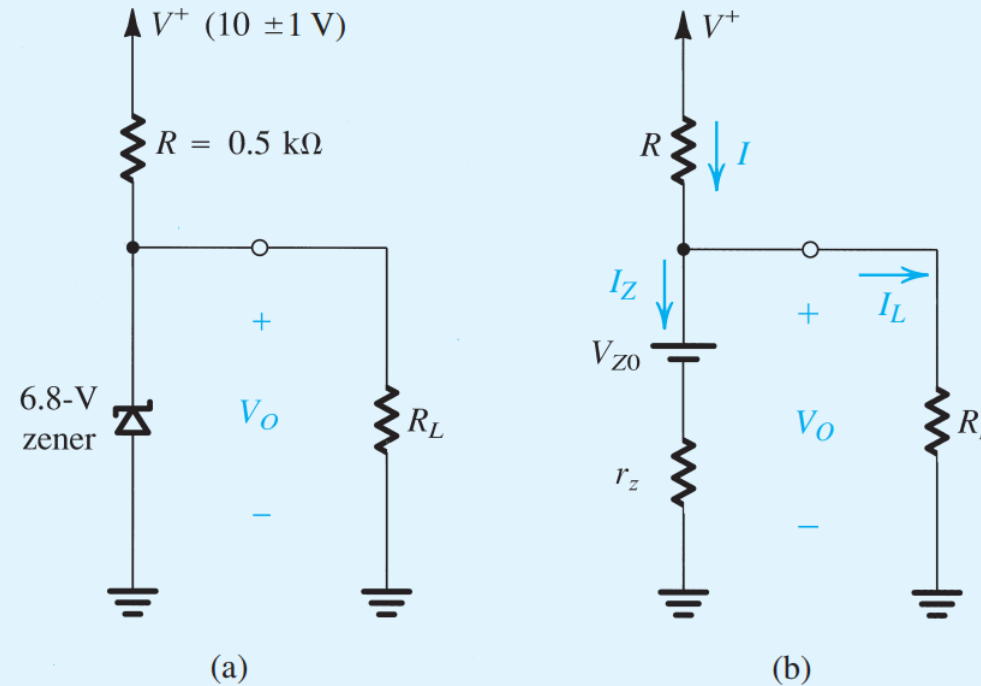


Figure 4.21 (a) Circuit for Example 4.7. (b) The circuit with the zener diode replaced with its equivalent circuit model.

Zener Diode

First we must determine the value of the parameter V_{z0} of the zener diode model. Substituting $V_z = 6.8$ V, $I_z = 5$ mA, and $r_z = 20$ Ω in Eq. (4.20) yields $V_{z0} = 6.7$ V. Figure 4.21(b) shows the circuit with the zener diode replaced with its model.

(a) With no load connected, the current through the zener is given by

$$\begin{aligned} I_z = I &= \frac{V^+ - V_{z0}}{R + r_z} \\ &= \frac{10 - 6.7}{0.5 + 0.02} = 6.35 \text{ mA} \end{aligned}$$

Thus,

$$\begin{aligned} V_o &= V_{z0} + I_z r_z \\ &= 6.7 + 6.35 \times 0.02 = 6.83 \text{ V} \end{aligned}$$

(b) For a ± 1 -V change in V^+ , the change in output voltage can be found from

$$\begin{aligned} \Delta V_o &= \Delta V^+ \frac{r_z}{R + r_z} \\ &= \pm 1 \times \frac{20}{500 + 20} = \pm 38.5 \text{ mV} \end{aligned}$$

Thus,

$$\text{Line regulation} = 38.5 \text{ mV/V}$$

Zener Diode

(c) When a load resistance R_L that draws a load current $I_L = 1 \text{ mA}$ is connected, the zener current will decrease by 1 mA. The corresponding change in zener voltage can be found from

$$\begin{aligned}\Delta V_o &= r_z \Delta I_z \\ &= 20 \times -1 = -20 \text{ mV}\end{aligned}$$

Thus the load regulation is

$$\text{Load regulation} \equiv \frac{\Delta V_o}{\Delta I_L} = -20 \text{ mV/mA}$$

(d) When a load resistance of $2 \text{ k}\Omega$ is connected, the load current will be approximately $6.8 \text{ V} / 2 \text{ k}\Omega = 3.4 \text{ mA}$. Thus the change in zener current will be $\Delta I_z = -3.4 \text{ mA}$, and the corresponding change in zener voltage (output voltage) will thus be

$$\begin{aligned}\Delta V_o &= r_z \Delta I_z \\ &= 20 \times -3.4 = -68 \text{ mV}\end{aligned}$$

Zener Diode

(e) An R_L of $0.5 \text{ k}\Omega$ would draw a load current of $6.8/0.5 = 13.6 \text{ mA}$. This is not possible, because the current I supplied through R is only 6.4 mA (for $V^+ = 10 \text{ V}$). Therefore, the zener must be cut off. If this is indeed the case, then V_o is determined by the voltage divider formed by R_L and R (Fig. 4.21a),

$$\begin{aligned} V_o &= V^+ \frac{R_L}{R + R_L} \\ &= 10 \frac{0.5}{0.5 + 0.5} = 5 \text{ V} \end{aligned}$$

Since this voltage is lower than the breakdown voltage of the zener, the diode is indeed no longer operating in the breakdown region.

(f) For the zener to be at the edge of the breakdown region, $I_Z = I_{ZK} = 0.2 \text{ mA}$ and $V_Z \simeq V_{ZK} \simeq 6.7 \text{ V}$. At this point the lowest (worst-case) current supplied through R is $(9 - 6.7)/0.5 = 4.6 \text{ mA}$, and thus the load current is $4.6 - 0.2 = 4.4 \text{ mA}$. The corresponding value of R_L is

$$R_L = \frac{6.7}{4.4} \simeq 1.5 \text{ k}\Omega$$

Half-Wave Rectifier

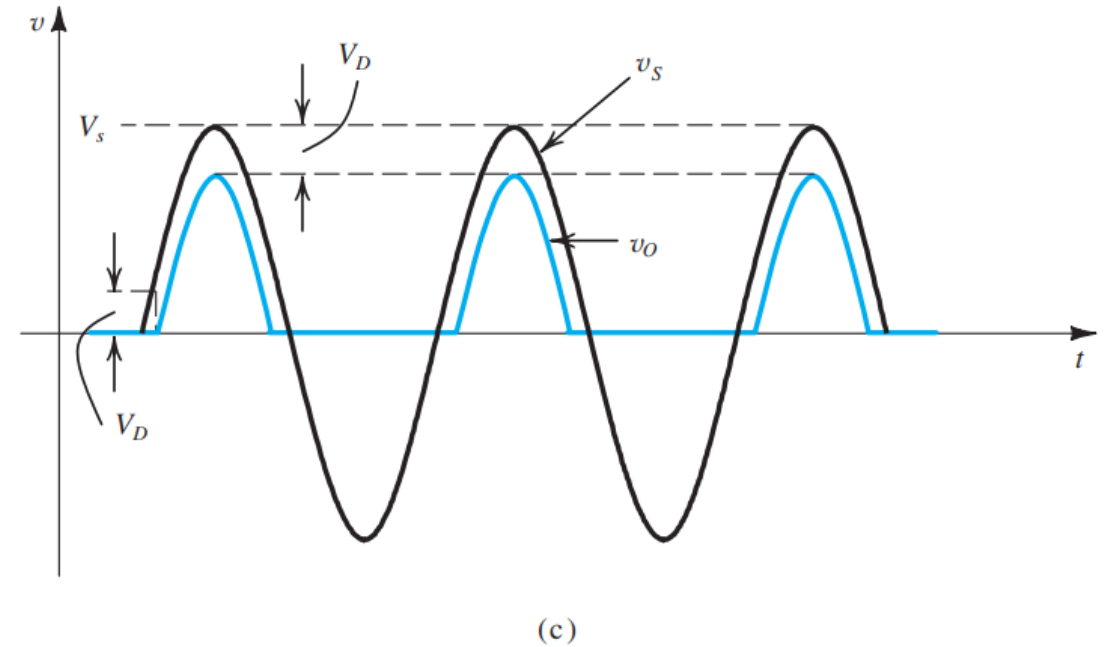
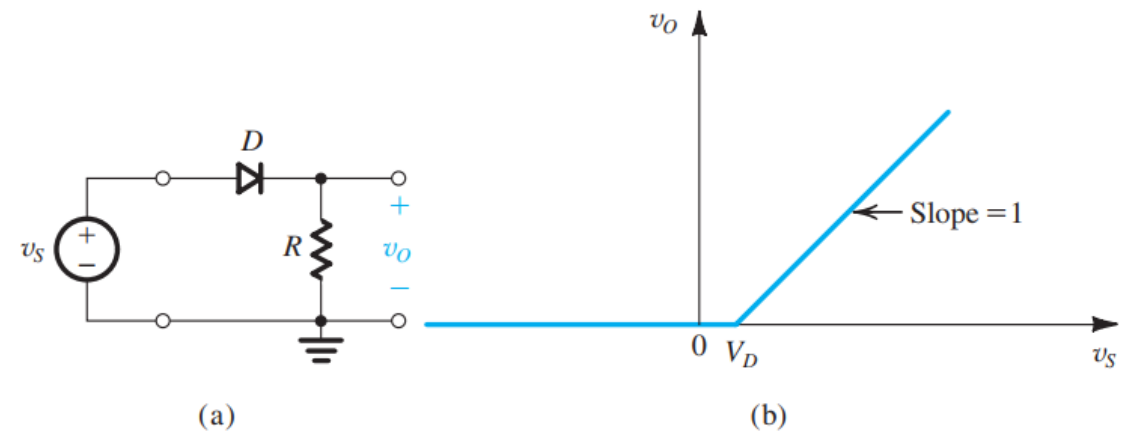
$$v_O = 0, \quad v_S < V_D$$

$$v_O = v_S - V_D, \quad v_S \geq V_D$$

$$\text{PIV} = V_S$$

- For the half-cycles during which the diode conducts, conduction begins at an angle $\theta = \sin^{-1}\left(\frac{V_D}{V_S}\right)$ and terminates at $(\pi - \theta)$, for a total conduction angle of $(\pi - 2\theta)$
- The average value (dc component) of v_O is

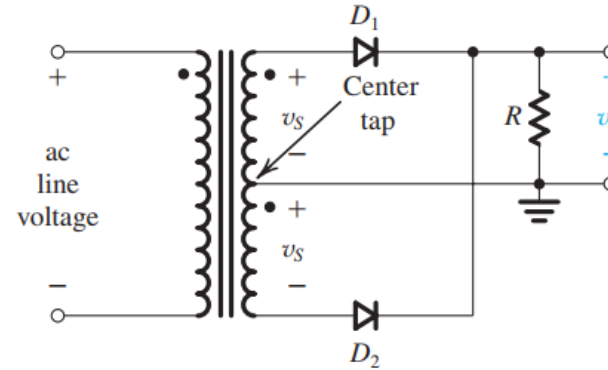
$$v_O \cong \frac{1}{\pi} V_S - \frac{V_D}{2}$$
- The peak diode current is $\frac{V_S - V_D}{R}$



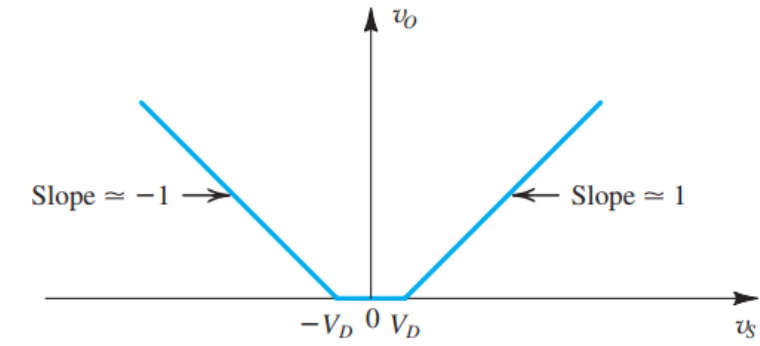
(a) Half-wave rectifier. (b) Transfer characteristic of the rectifier circuit. (c) Input and output waveforms.

Center-Tapped Full-Wave Rectifier

$$\text{PIV} = 2V_s - V_D$$

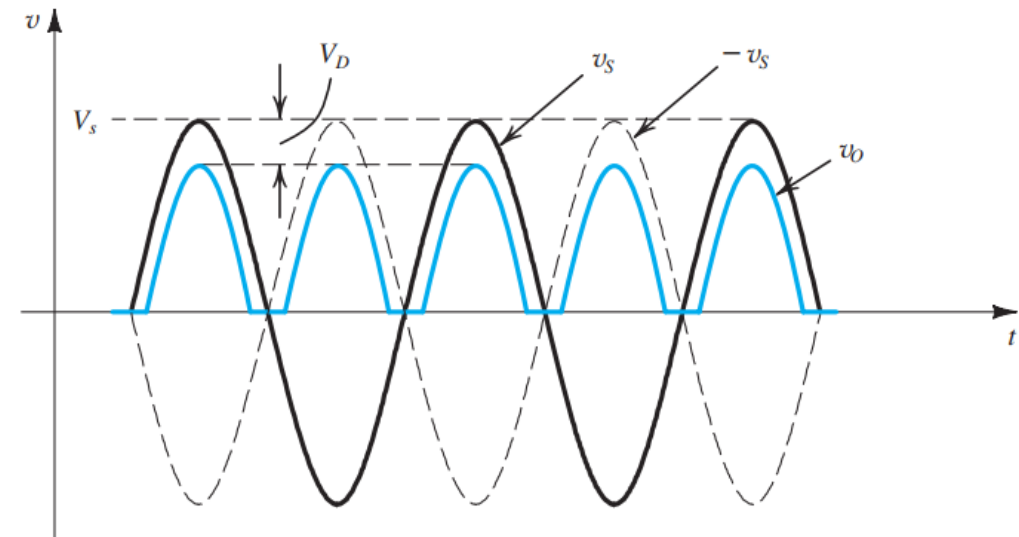


(a)



(b)

- The output is zero for an angle of $2 \sin^{-1} \left(\frac{V_D}{V_s} \right)$ centered around the zero-crossing points of the sine-wave input.
- The average value (dc component) of v_O is $v_O \cong \frac{2}{\pi} V_s - V_D$
- The peak current through each diode is $\frac{V_s - V_D}{R}$



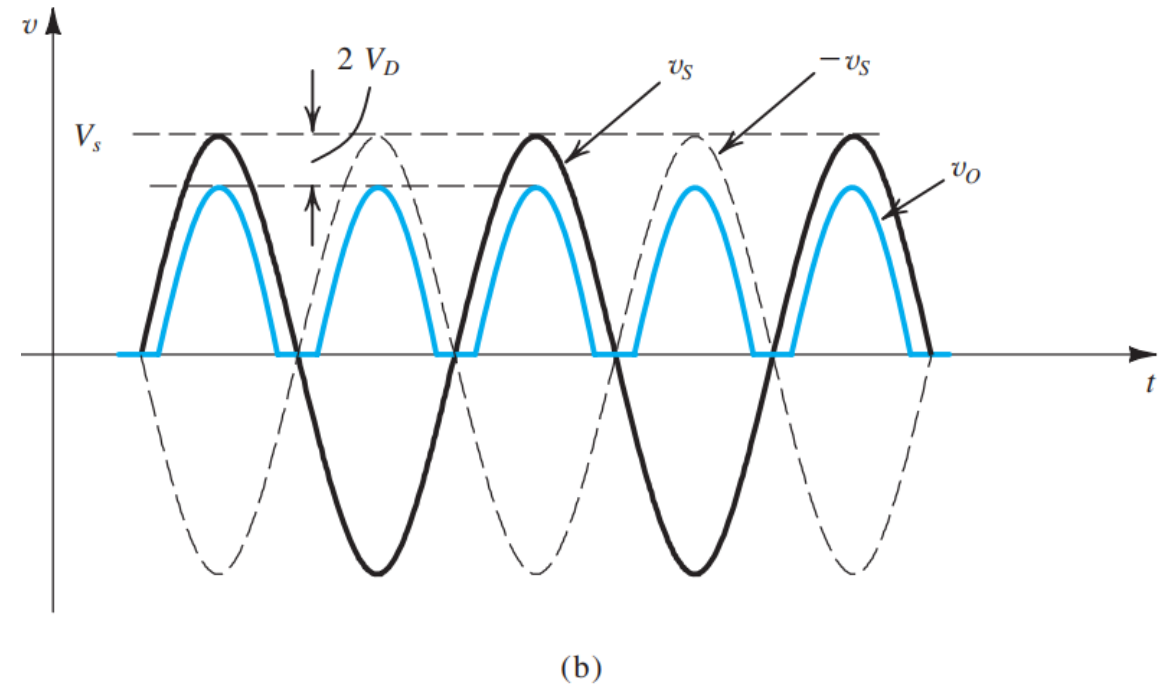
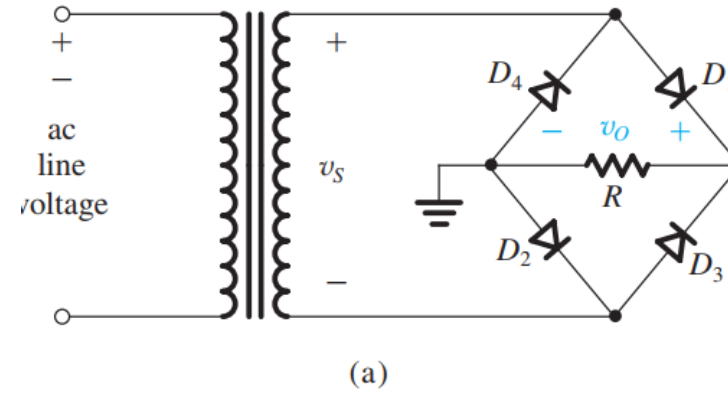
(c)

Full-wave rectifier utilizing a transformer with a center-tapped secondary winding: (a) circuit; (b) transfer characteristic assuming a constant-voltage-drop model for the diodes; (c) input and output waveforms.

Bridge Full-Wave Rectifier

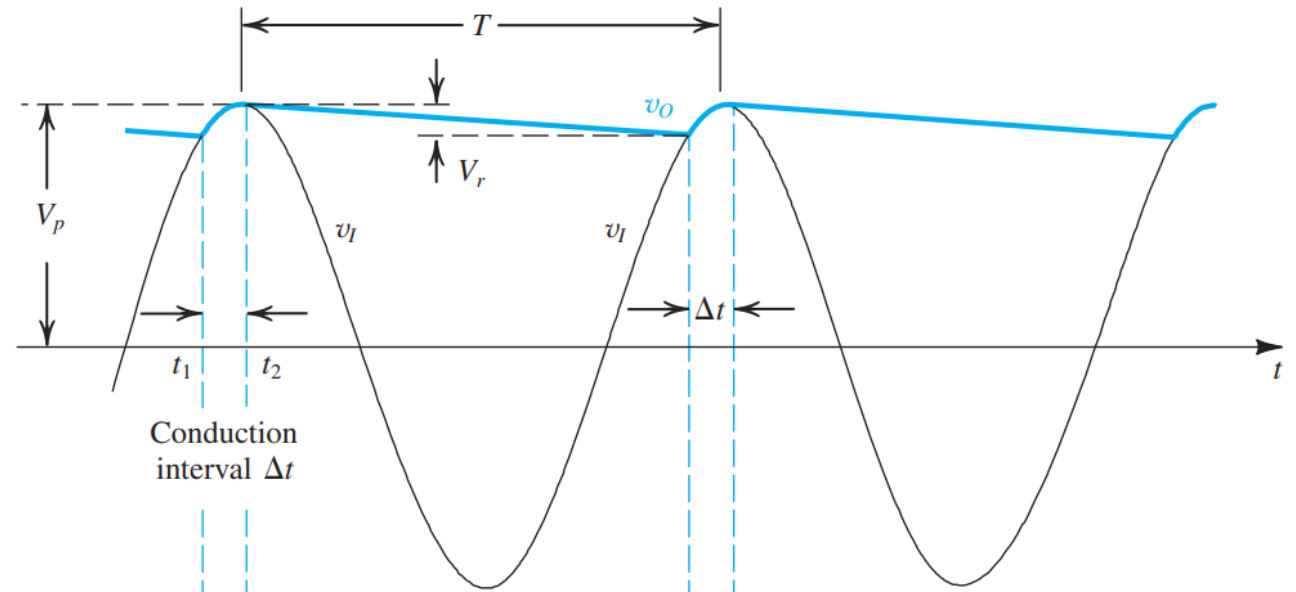
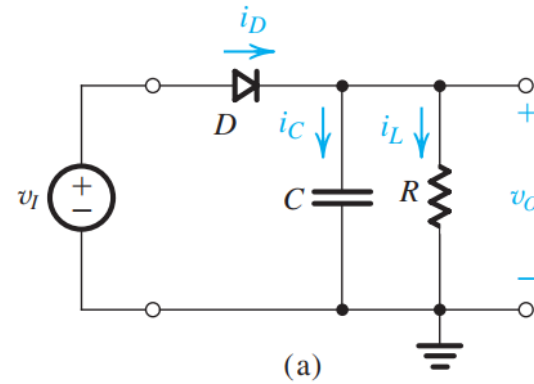
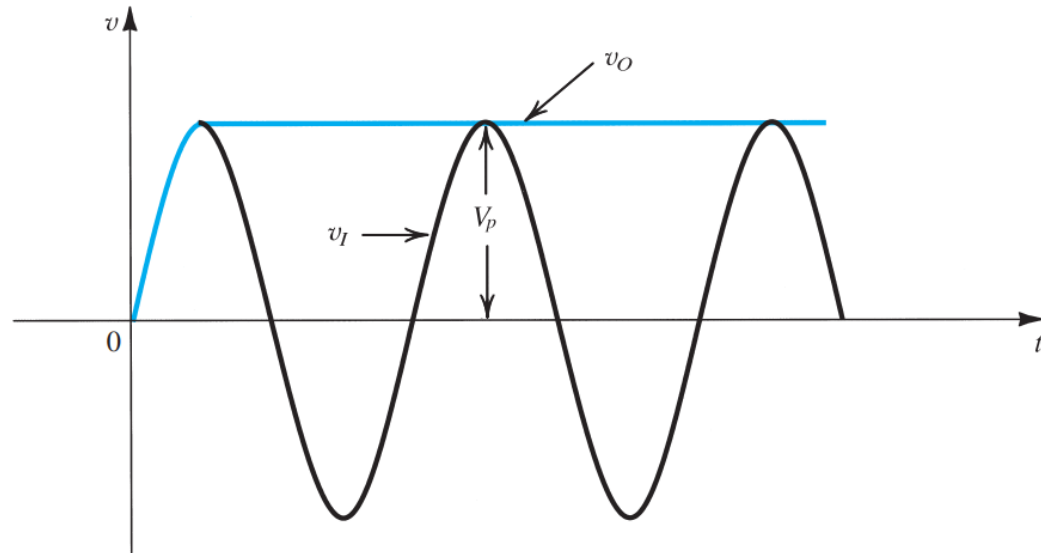
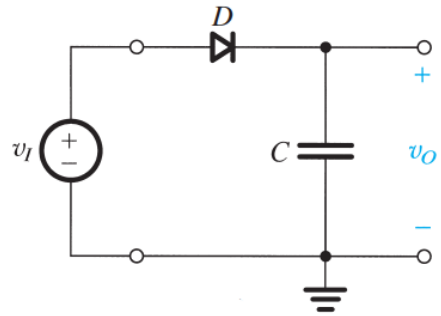
$$\text{PIV} = V_s - 2V_D + V_D = V_s - V_D$$

- The average (or dc component) of the output voltage is $v_O \cong \frac{2}{\pi} V_s - 2V_D$
- The peak current through each diode is $\frac{V_s - 2V_D}{R}$

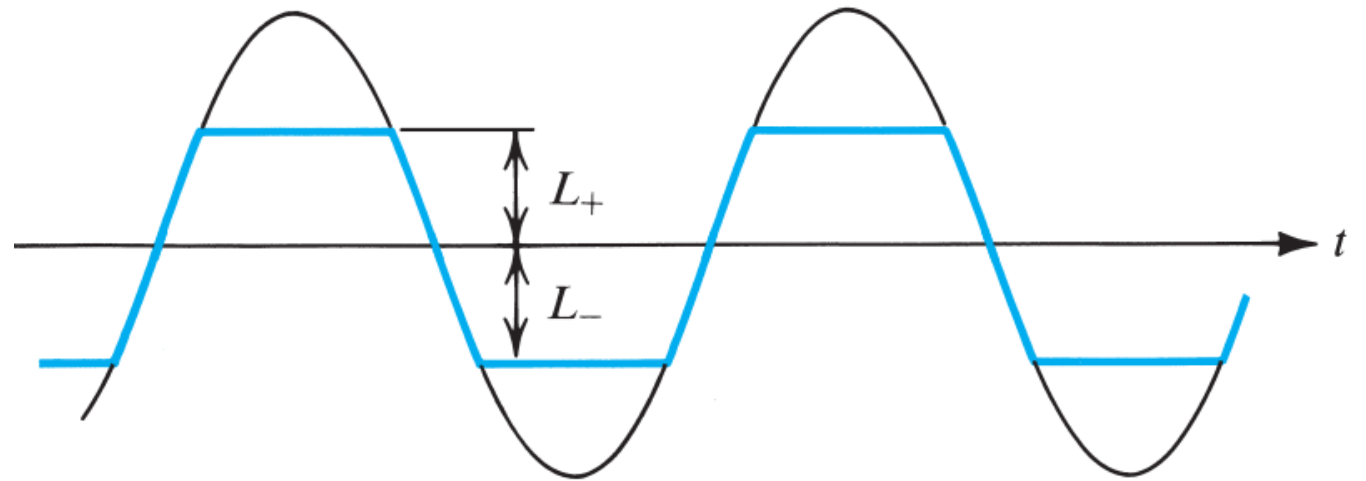
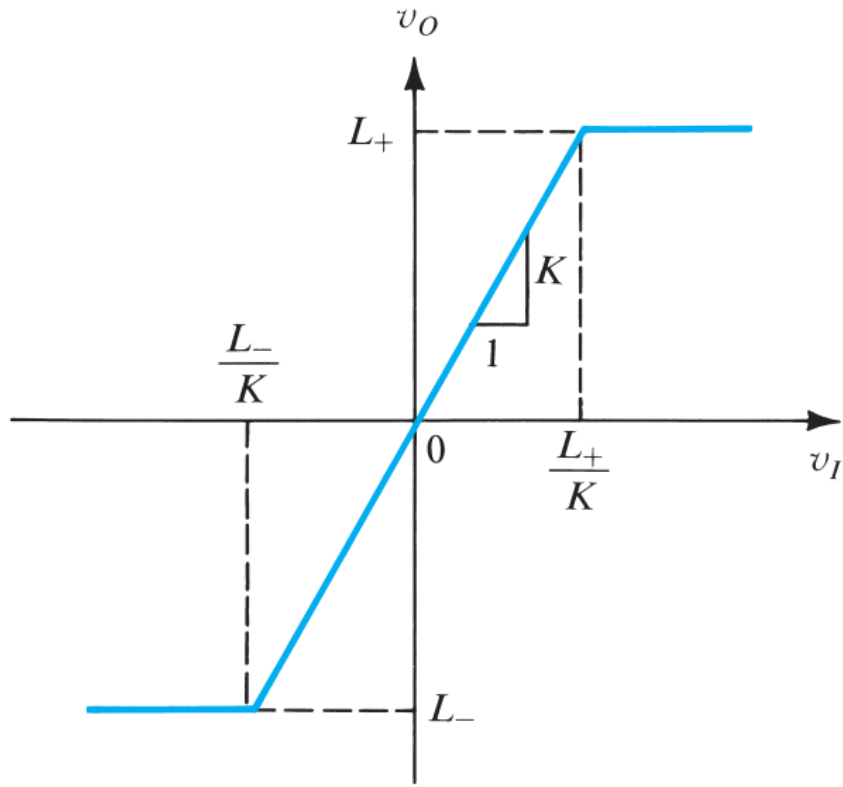


The bridge rectifier: (a) circuit; (b) input and output waveforms.

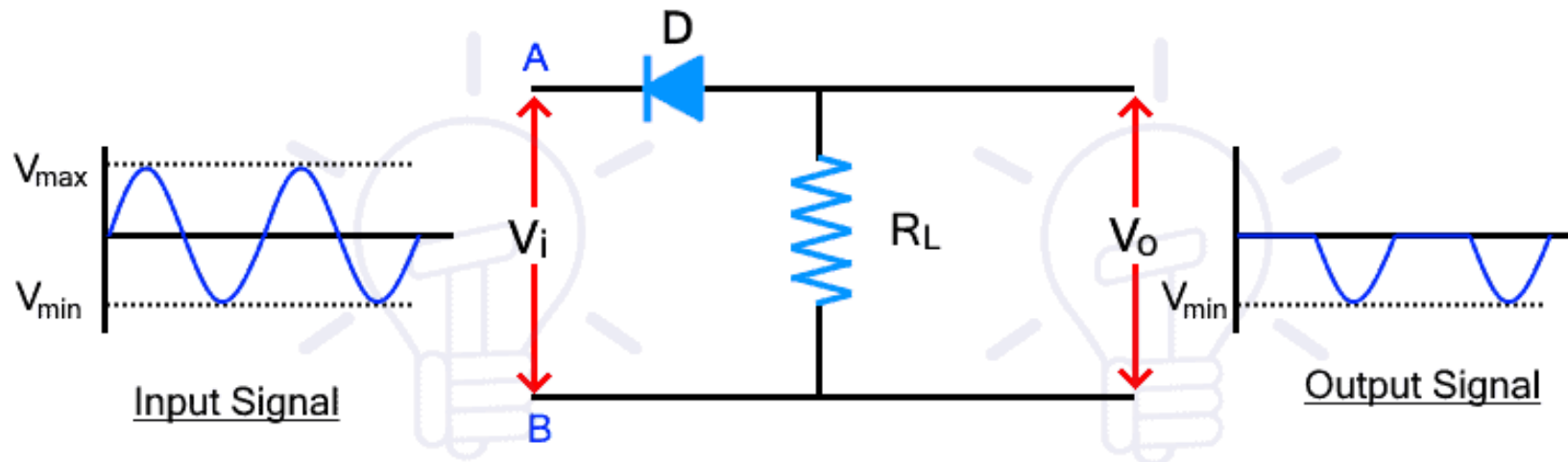
The Peak Rectifier



Clipper Circuit

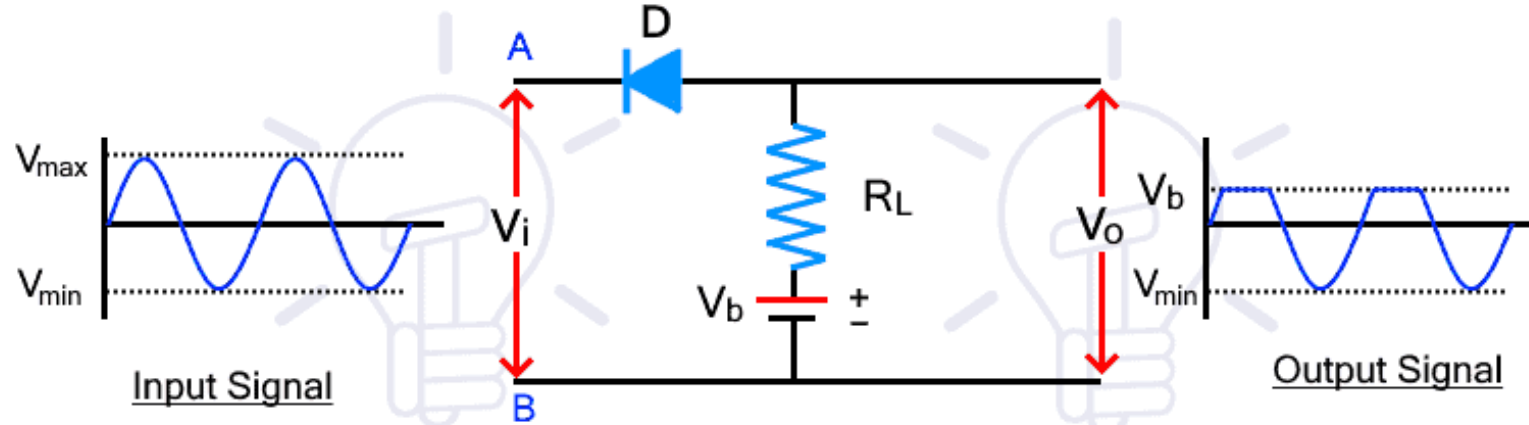


Clipper Circuit

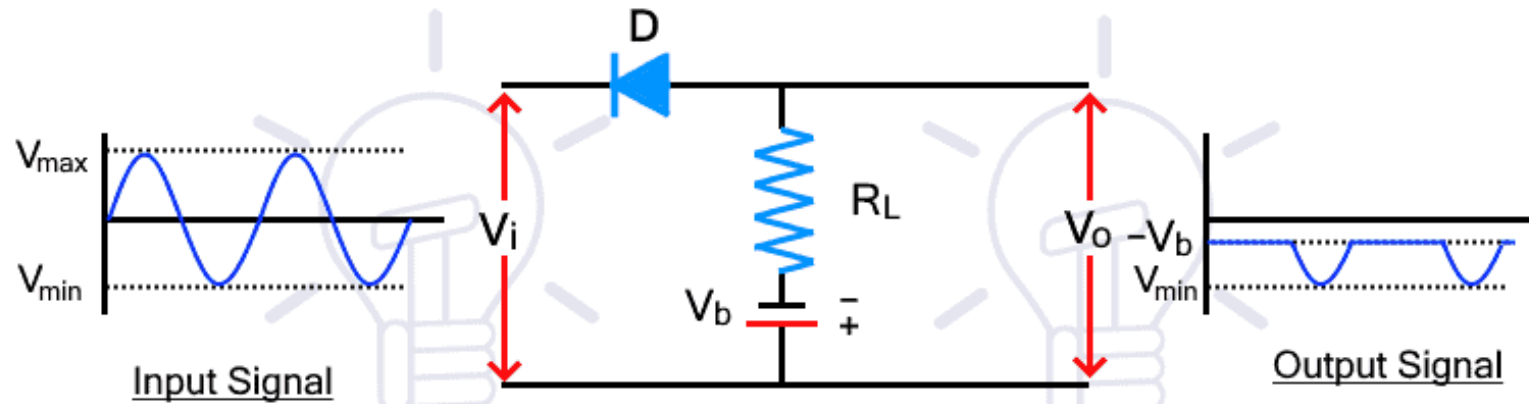


Series Positive Clipper

Clipper Circuit

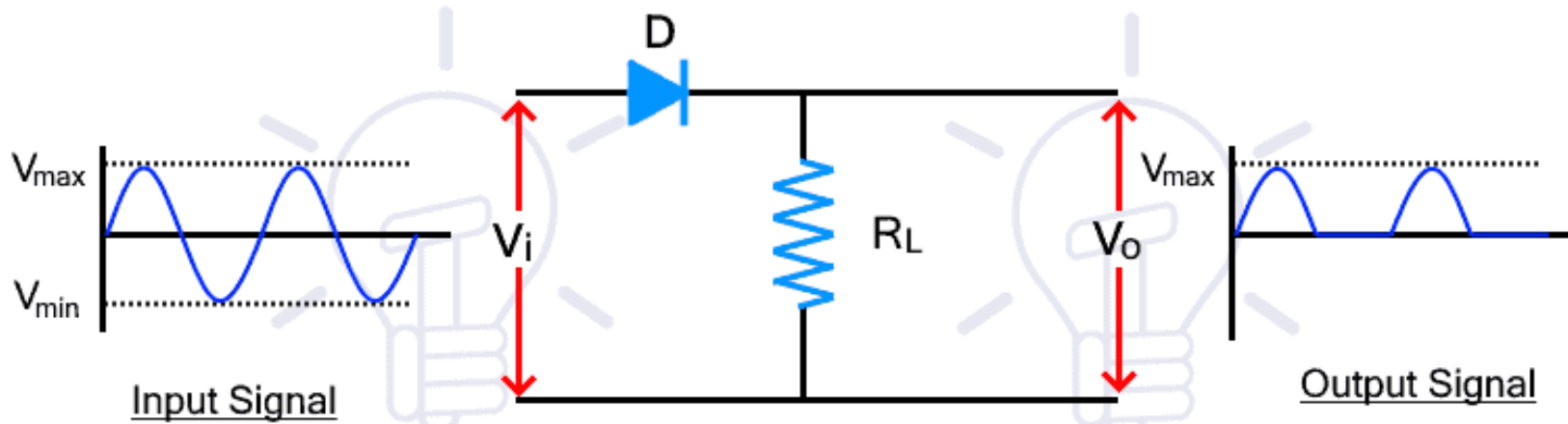


Series Positive Clipper with Positive Bias



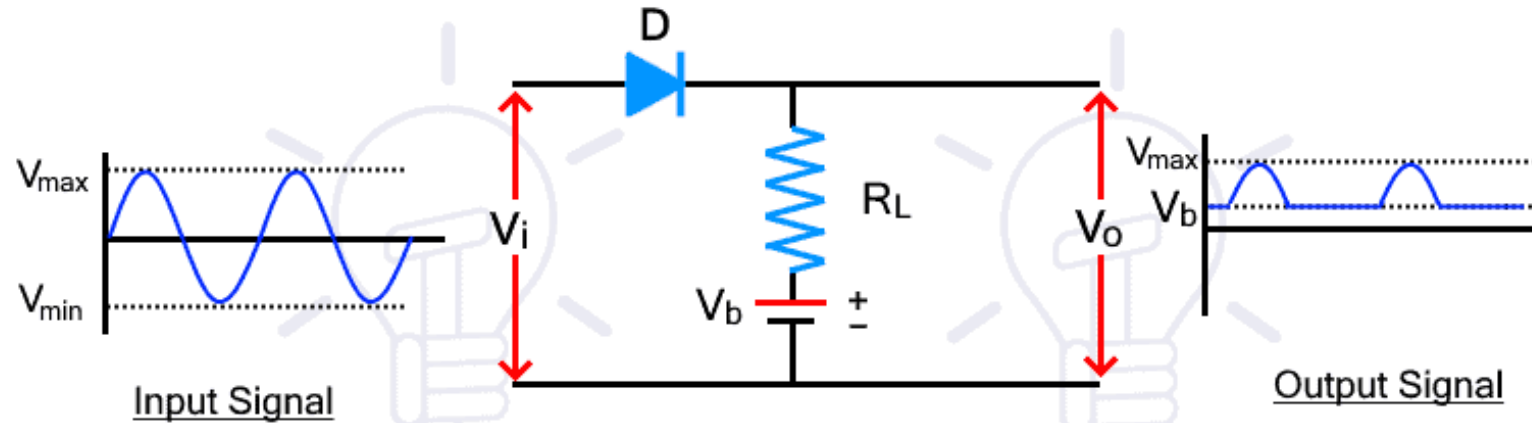
Series Positive Clipper with Negative Bias

Clipper Circuit

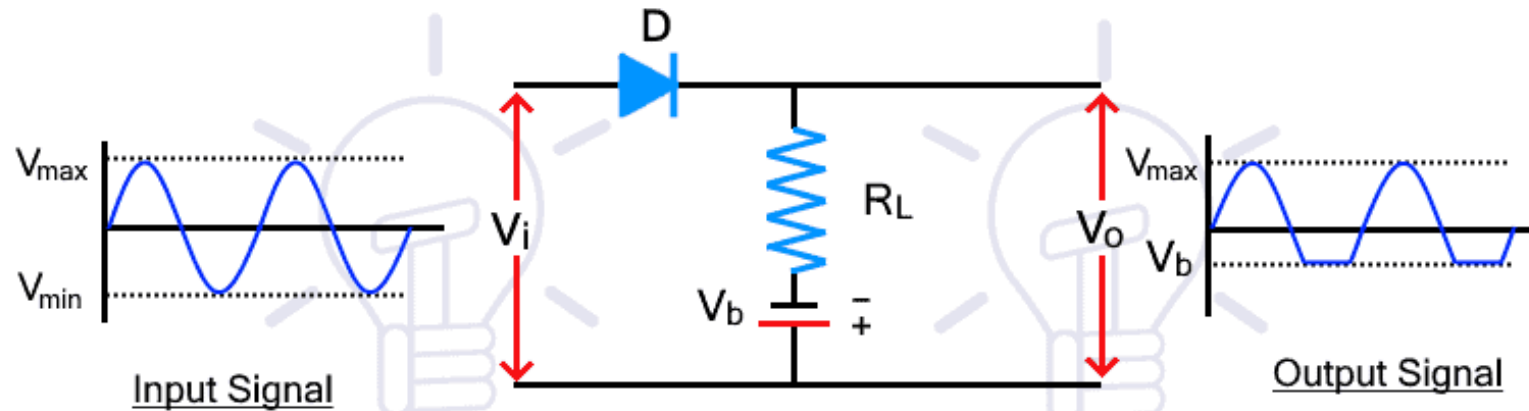


Series Negative Clipper

Clipper Circuit

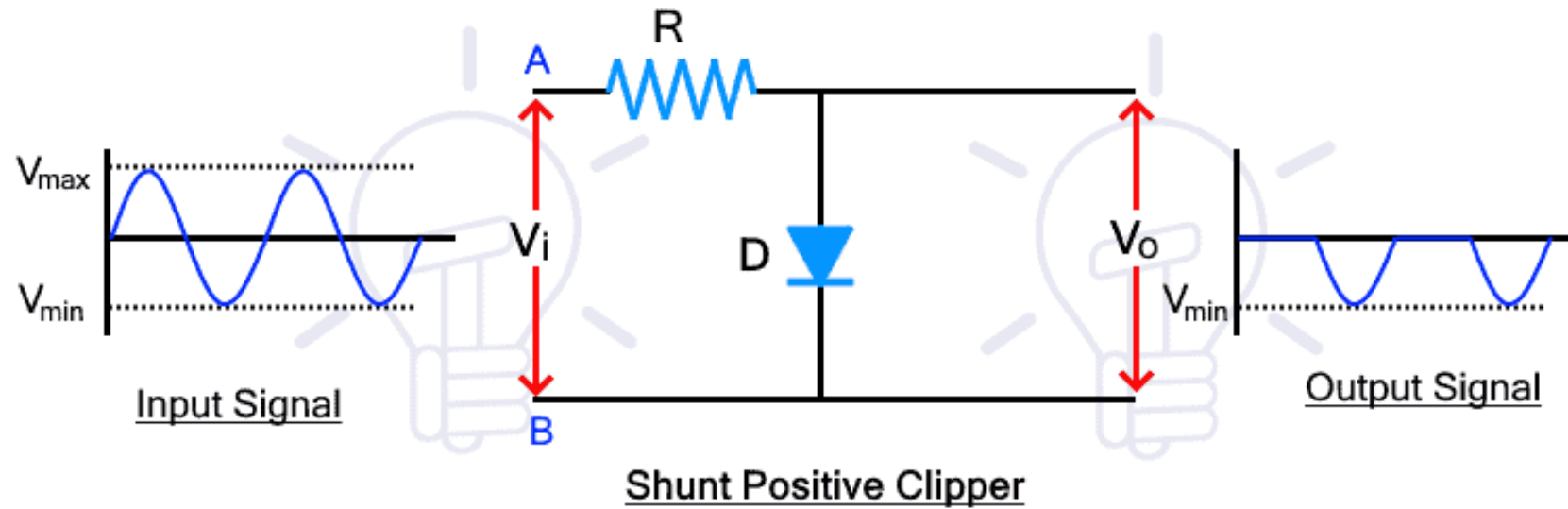


Series Negative Clipper with Positive Bias

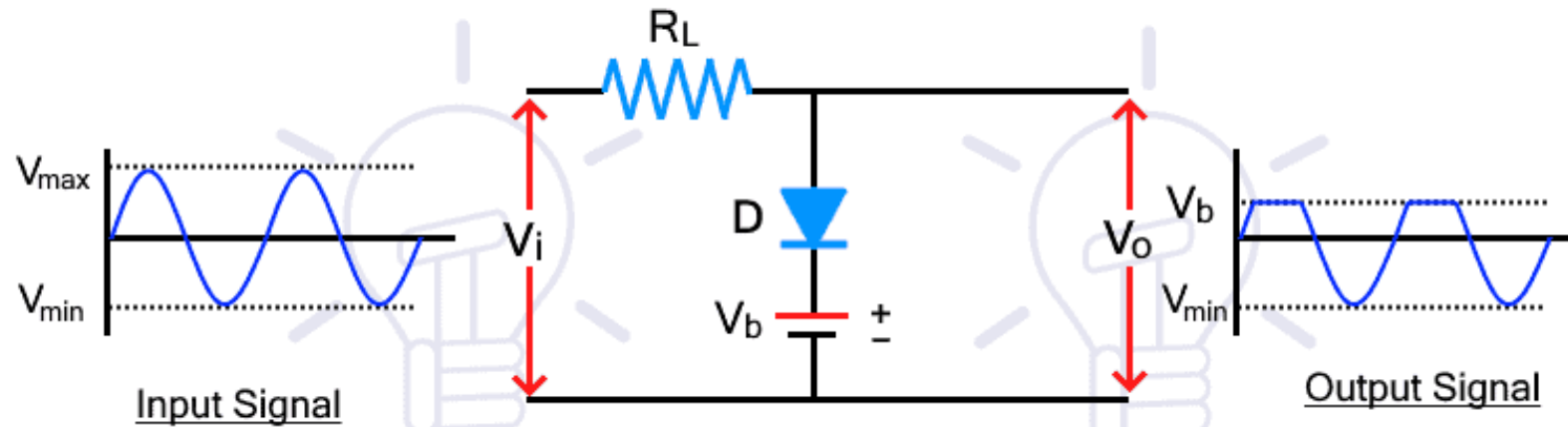


Series Negative Clipper with Negative Bias

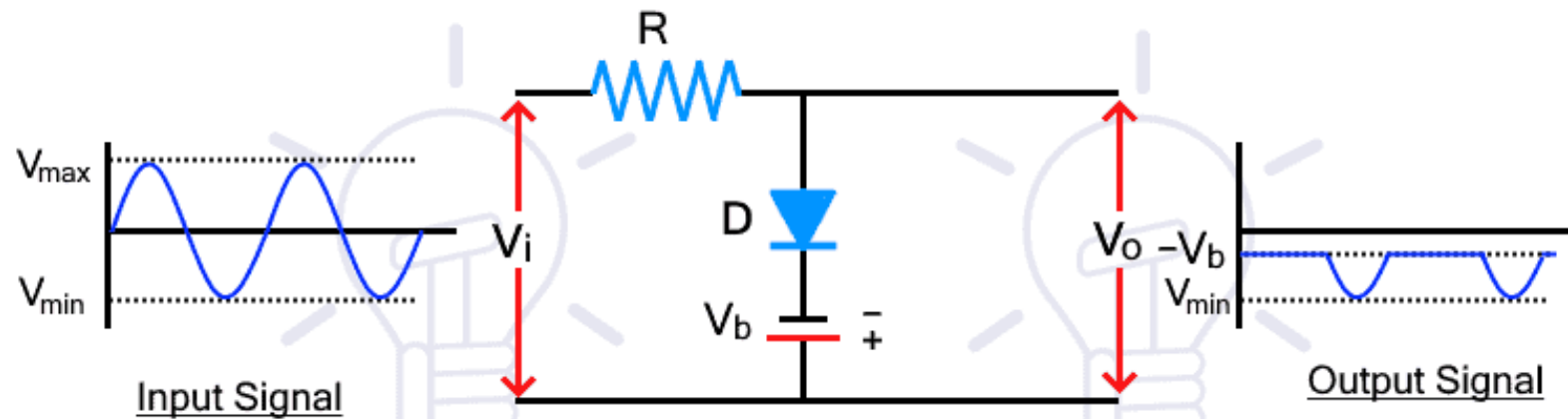
Clipper Circuit



Clipper Circuit

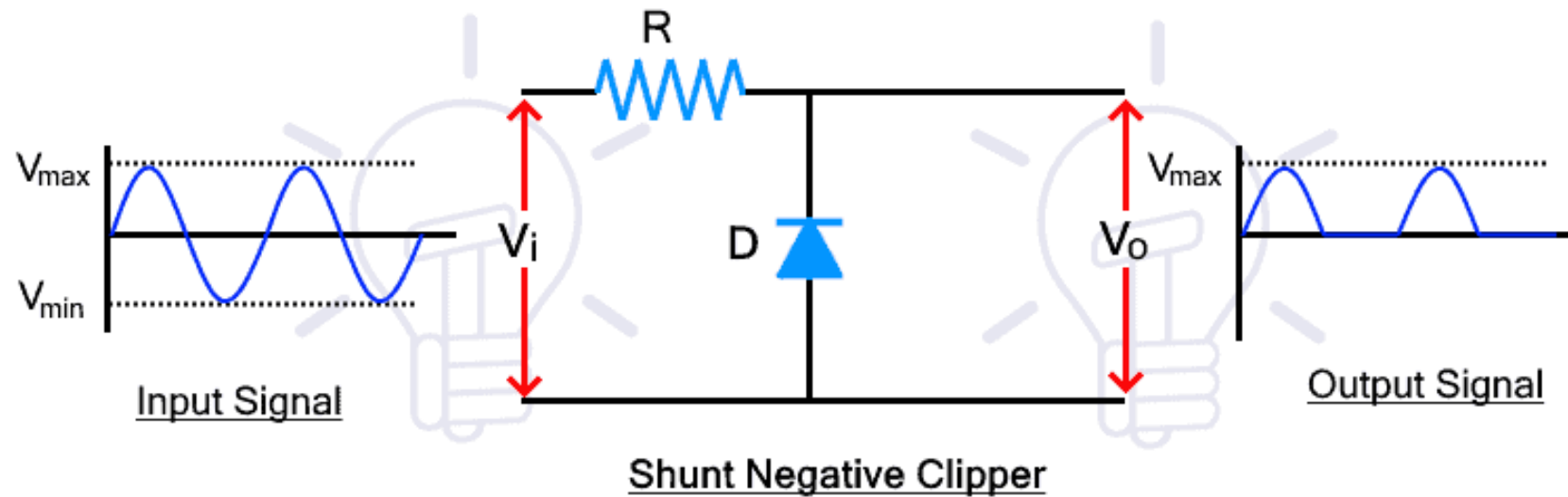


Shunt Positive Clipper with Positive Bias

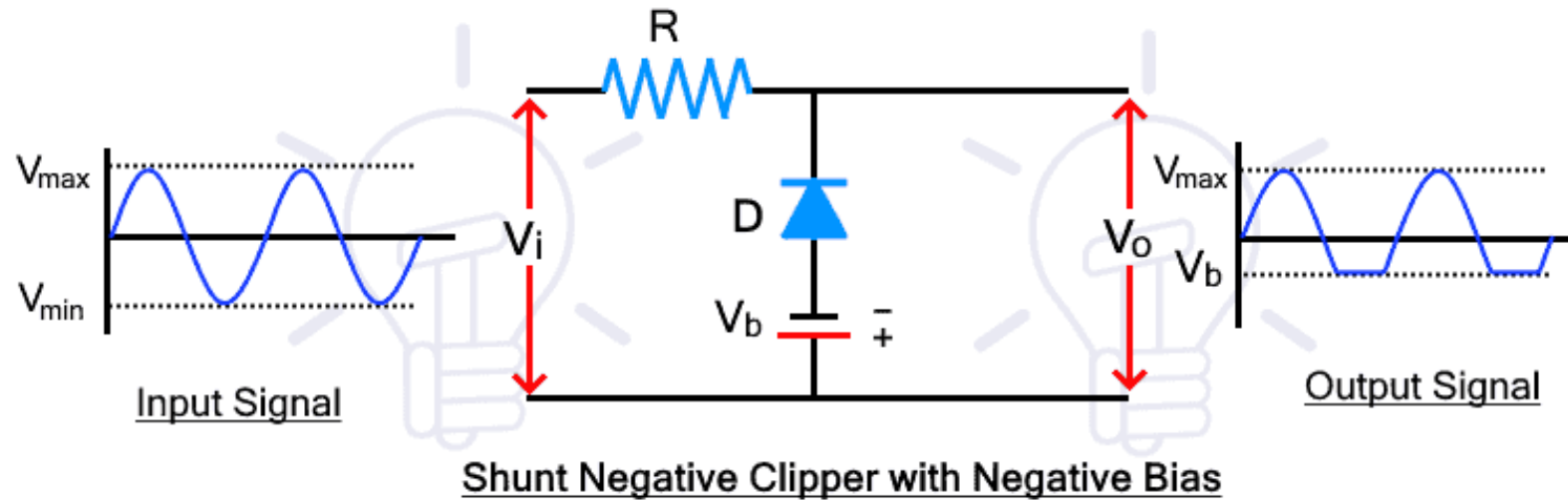
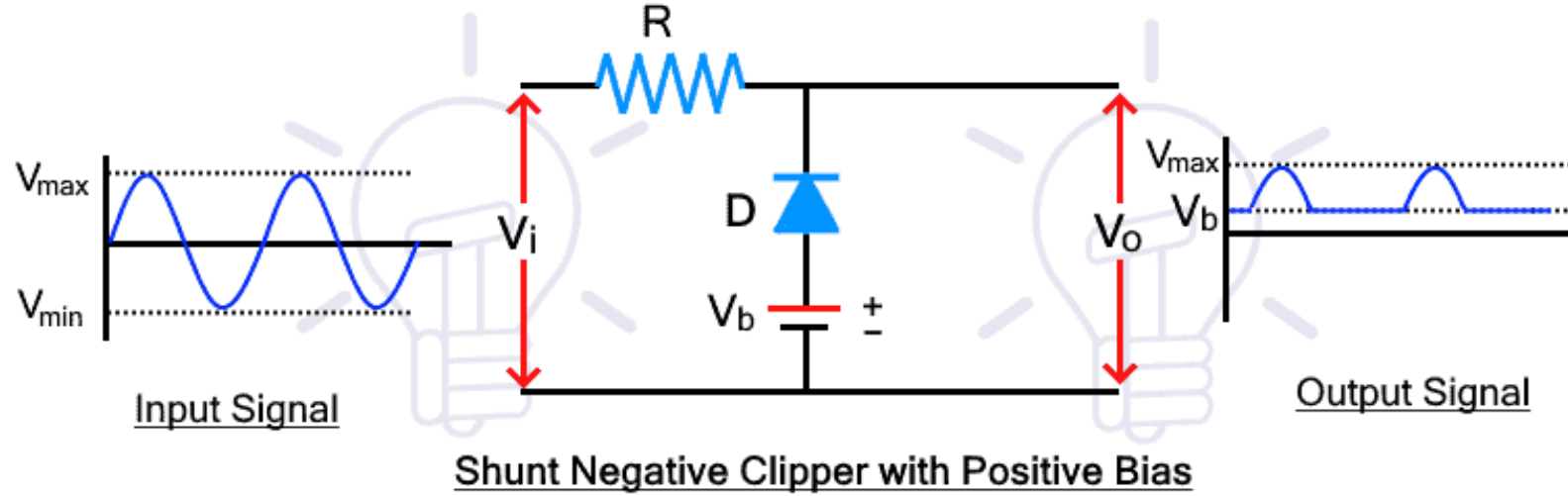


Shunt Positive Clipper with Negative Bias

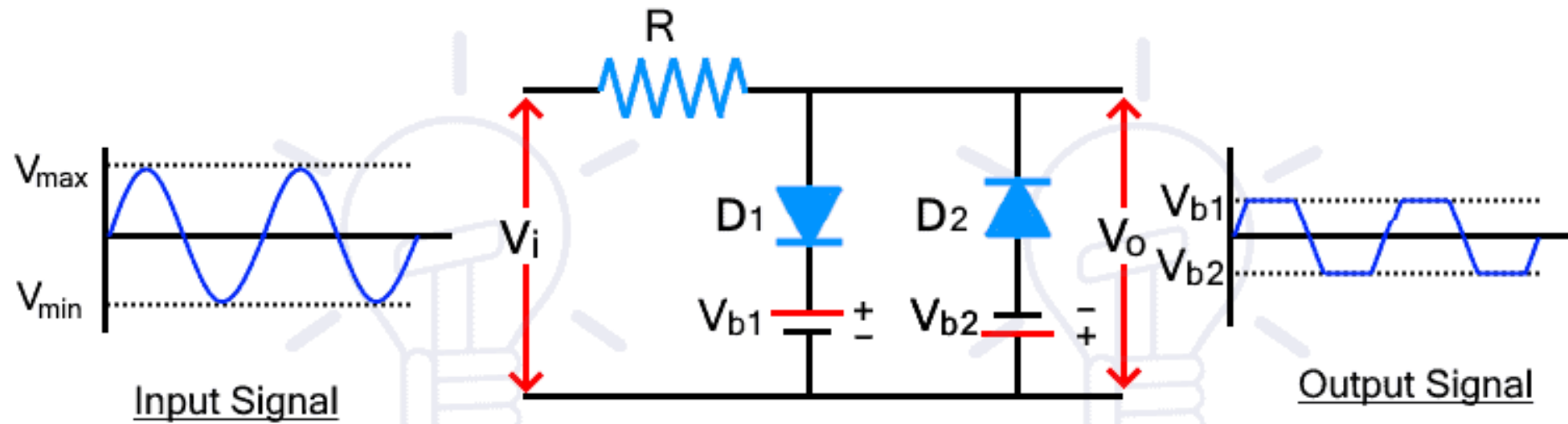
Clipper Circuit



Clipper Circuit

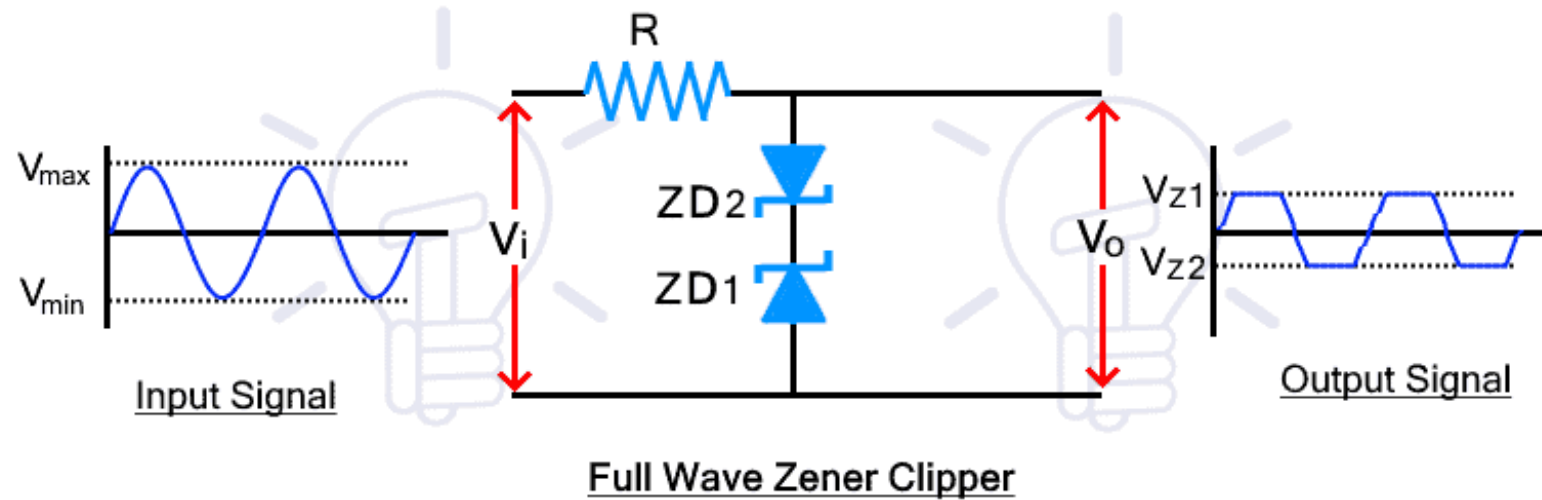
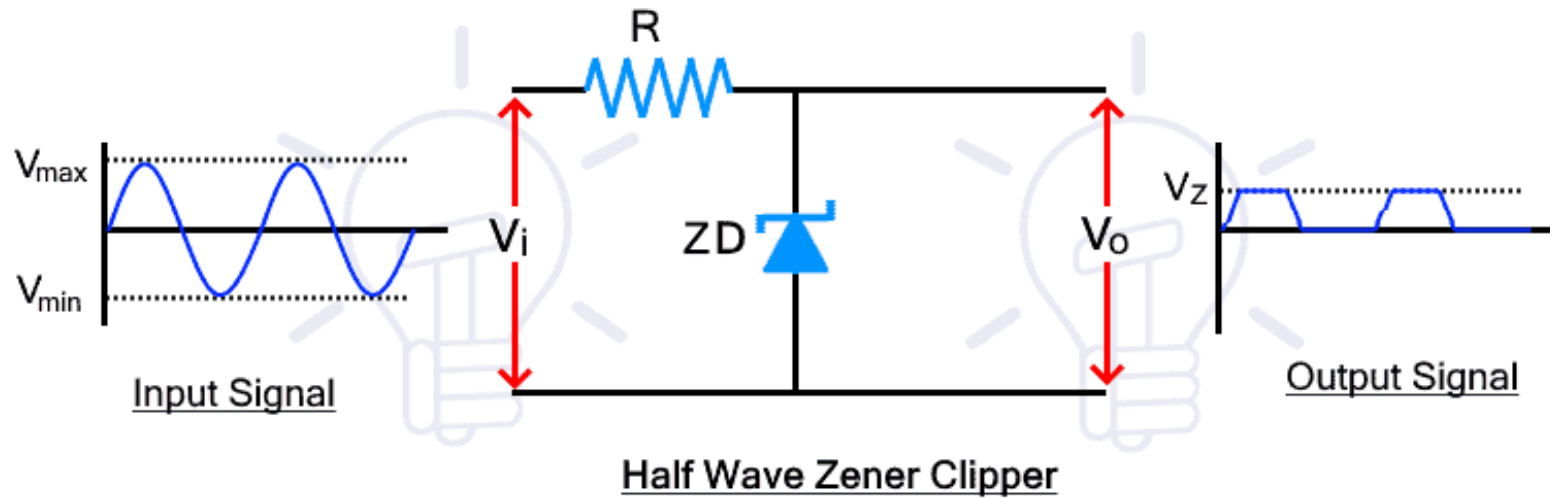


Clipper Circuit

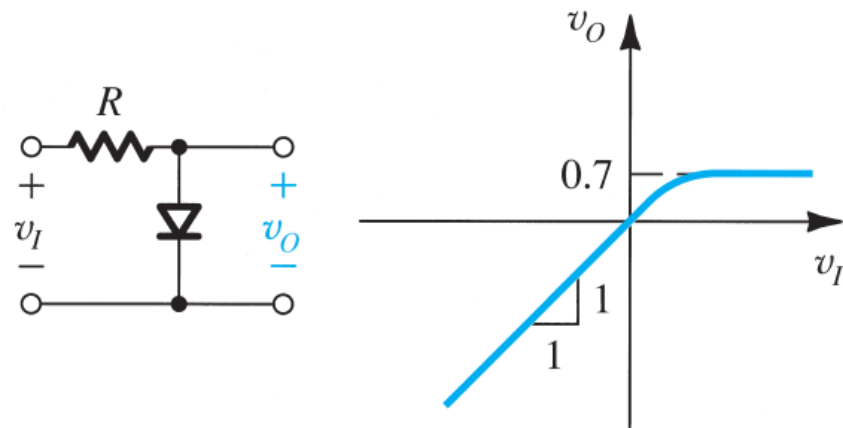


Dual / Combination Clipper

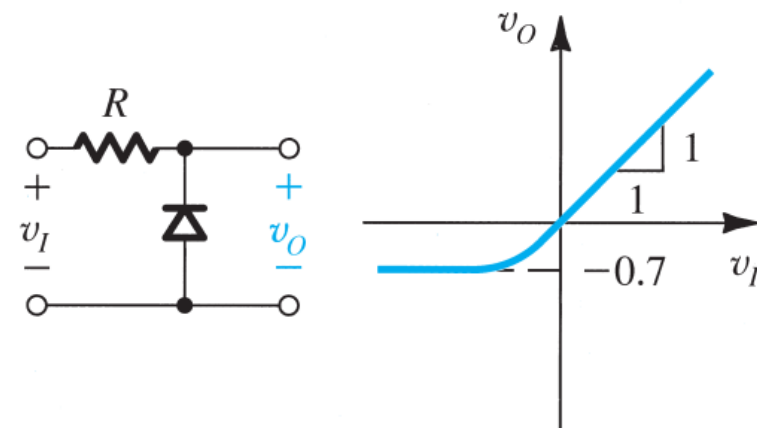
Clipper Circuit



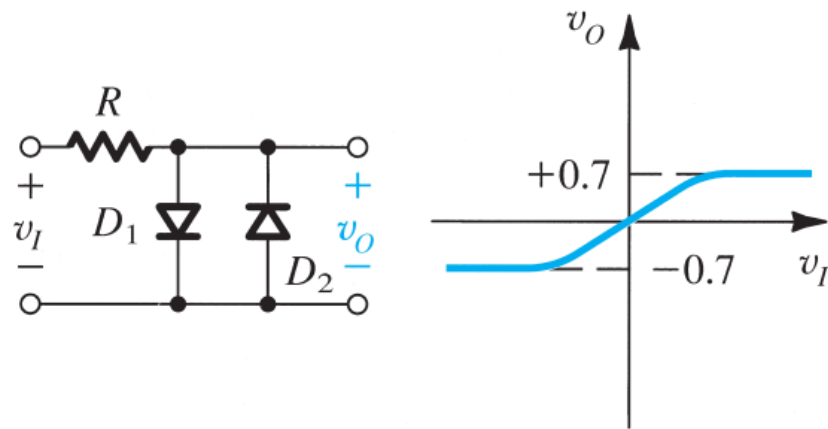
Clipper Circuit



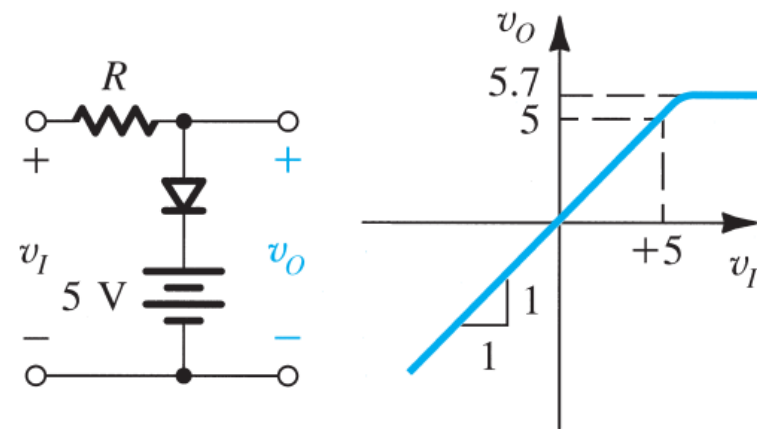
(a)



(b)



(c)



(d)

Clipper Circuit

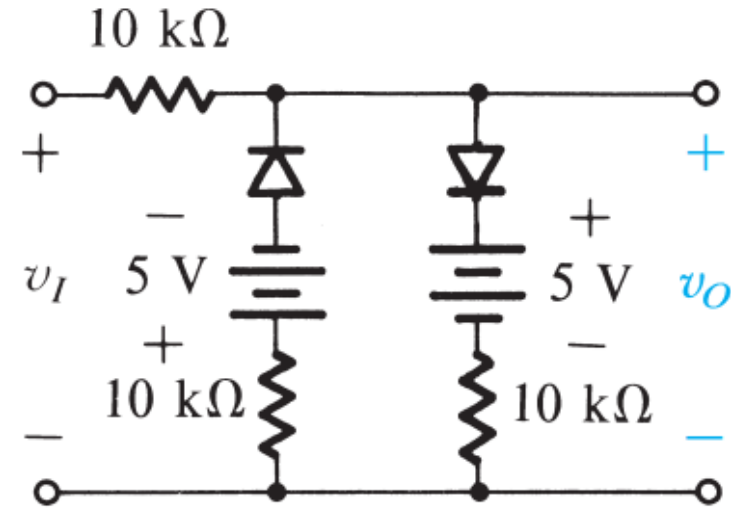
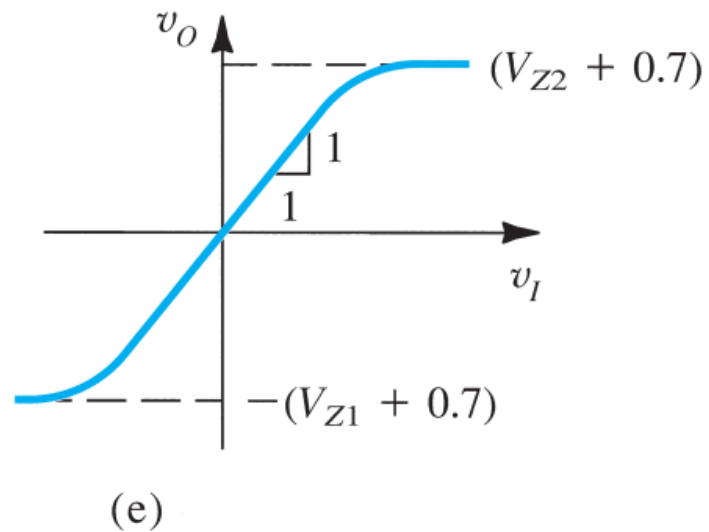
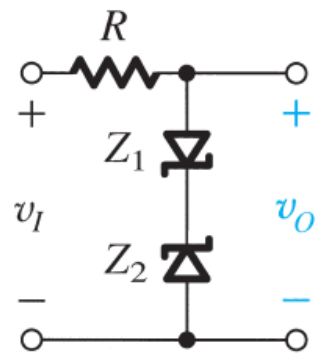


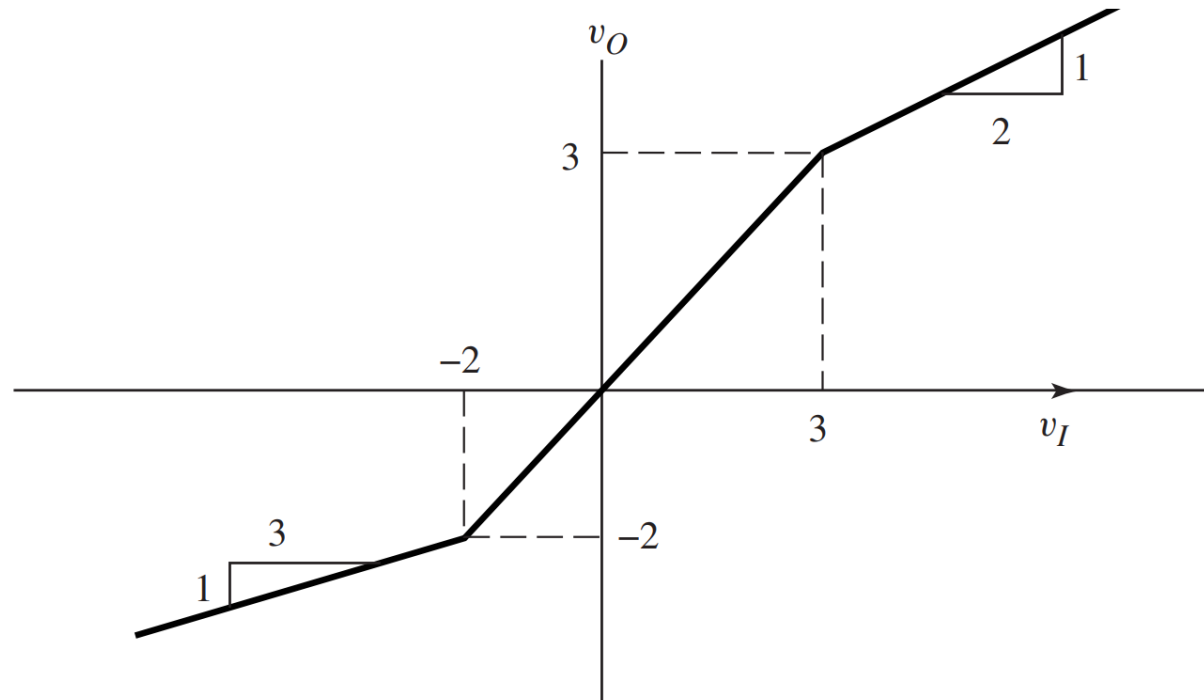
Figure E4.26

Ans.

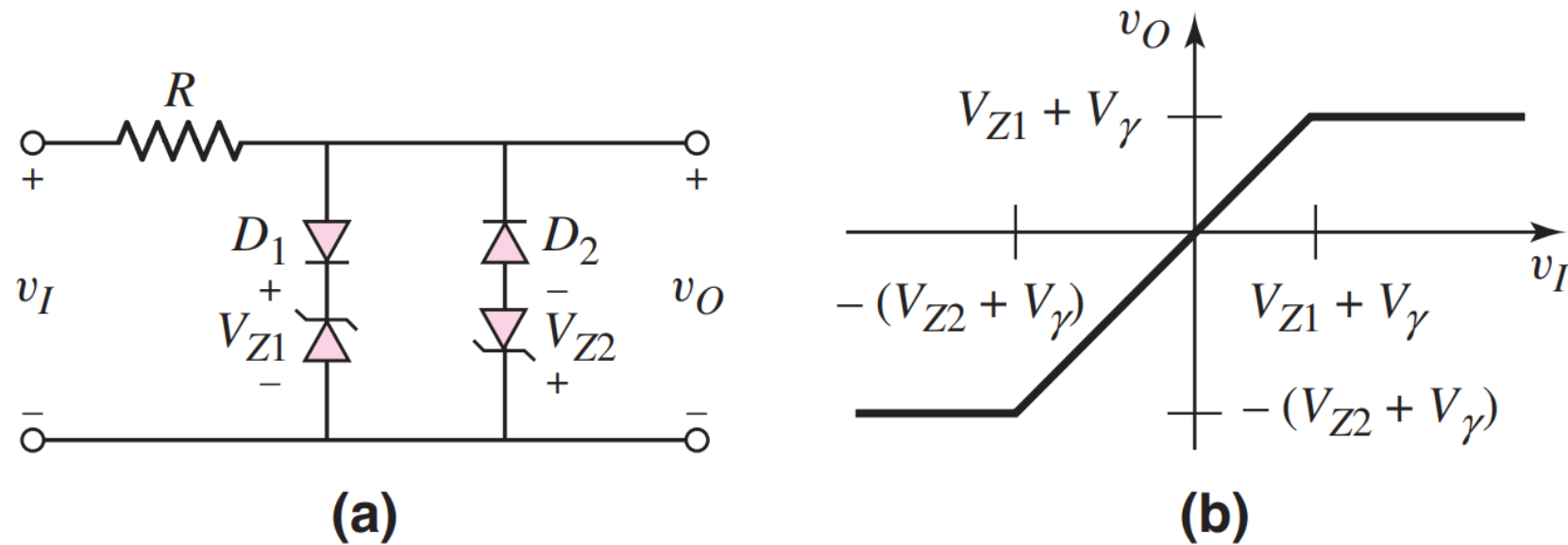
$v_O = v_I$	for $-5 \leq v_I \leq +5$
$v_O = \frac{1}{2}v_I - 2.5$	for $v_I \leq -5$
$v_O = \frac{1}{2}v_I + 2.5$	for $v_I \geq +5$

Clipper Circuit

Q. Design a parallel-based clipper circuit that will yield the voltage transfer characteristics as shown below. Assume a diode cut-in voltage 0.7 V.

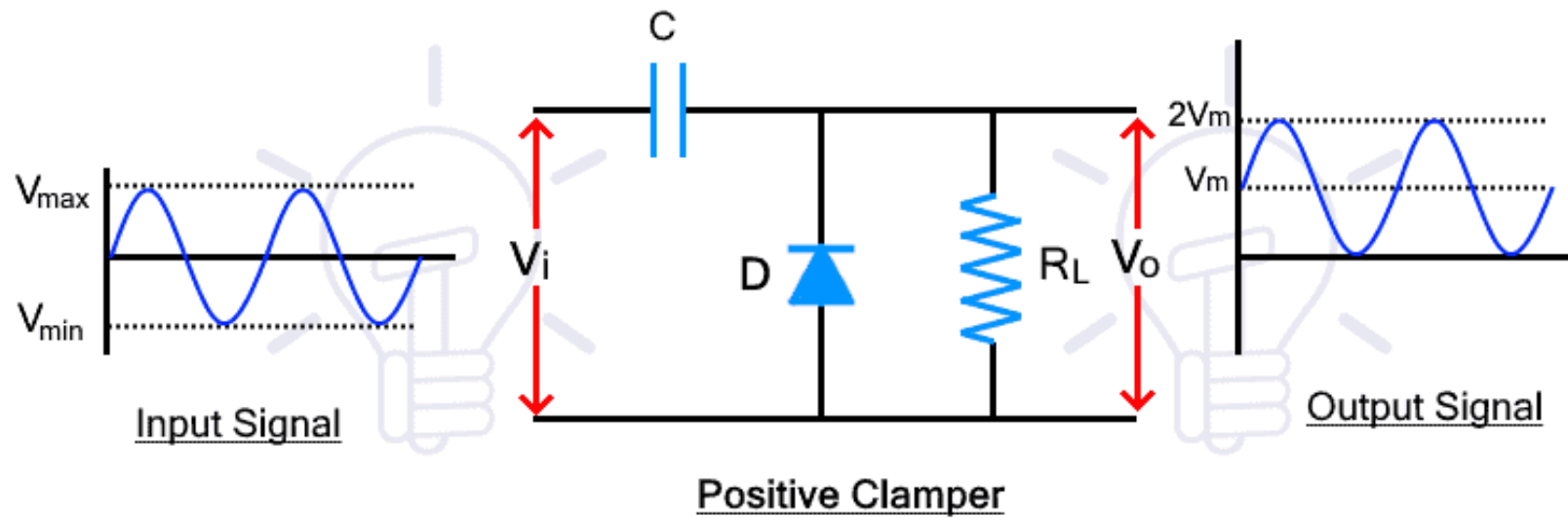


Clipper Circuit



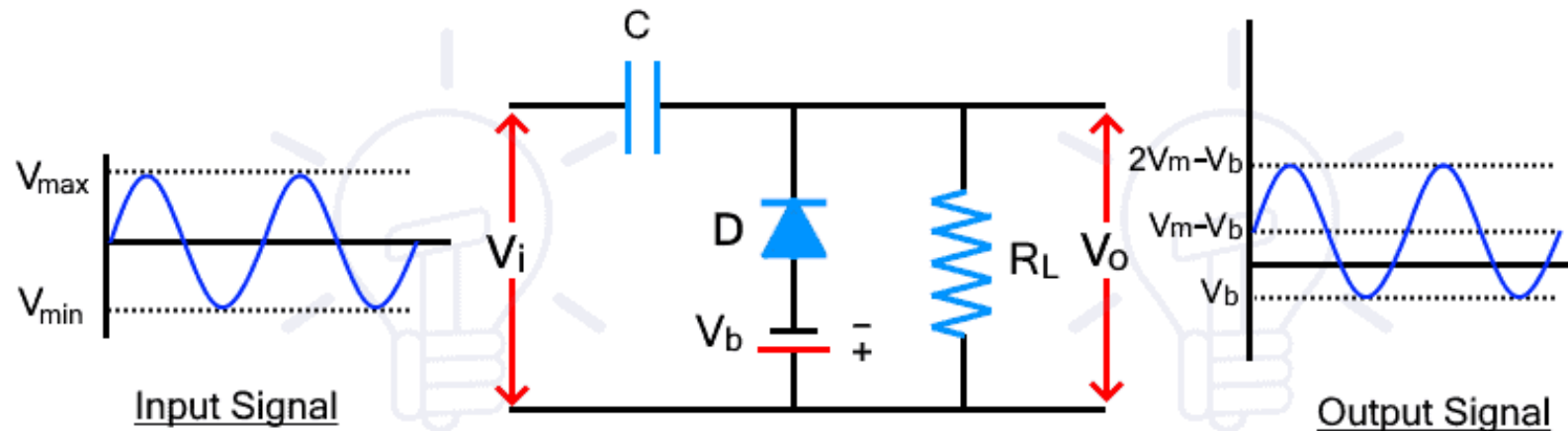
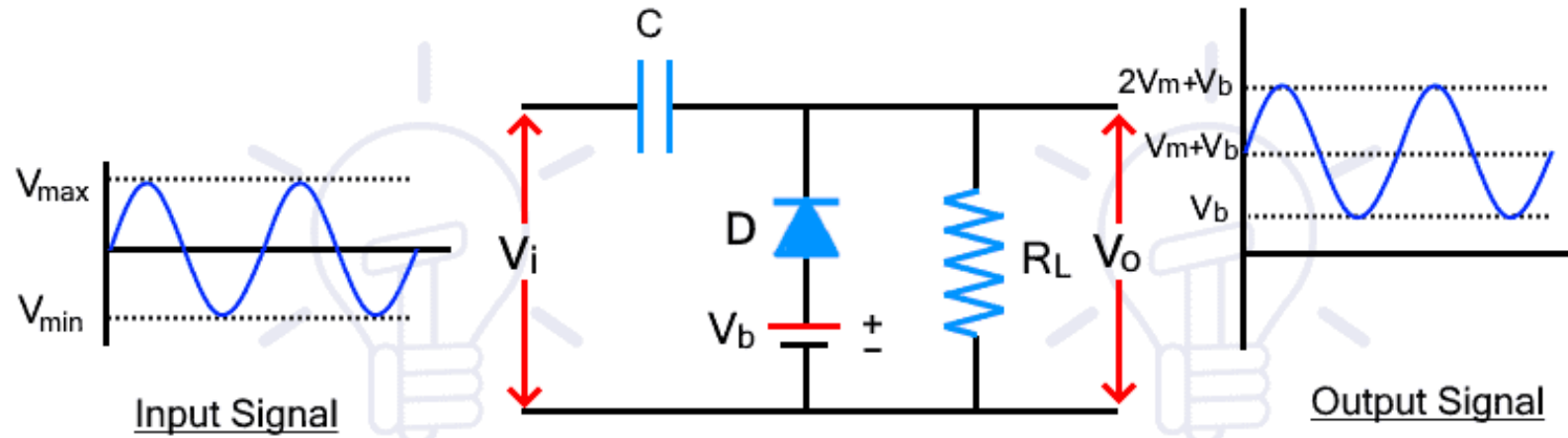
(a) Parallel-based clipper circuit using Zener diodes; (b) voltage transfer characteristics

Clamper Circuit



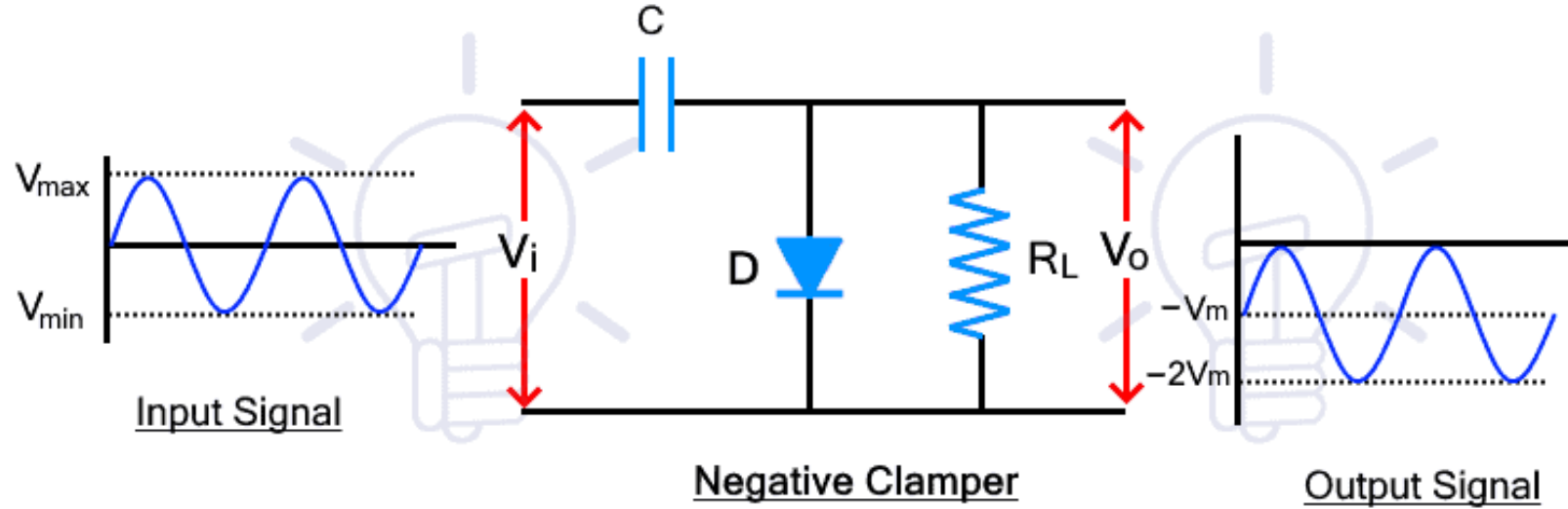
$$RC \gg T$$

Clamper Circuit



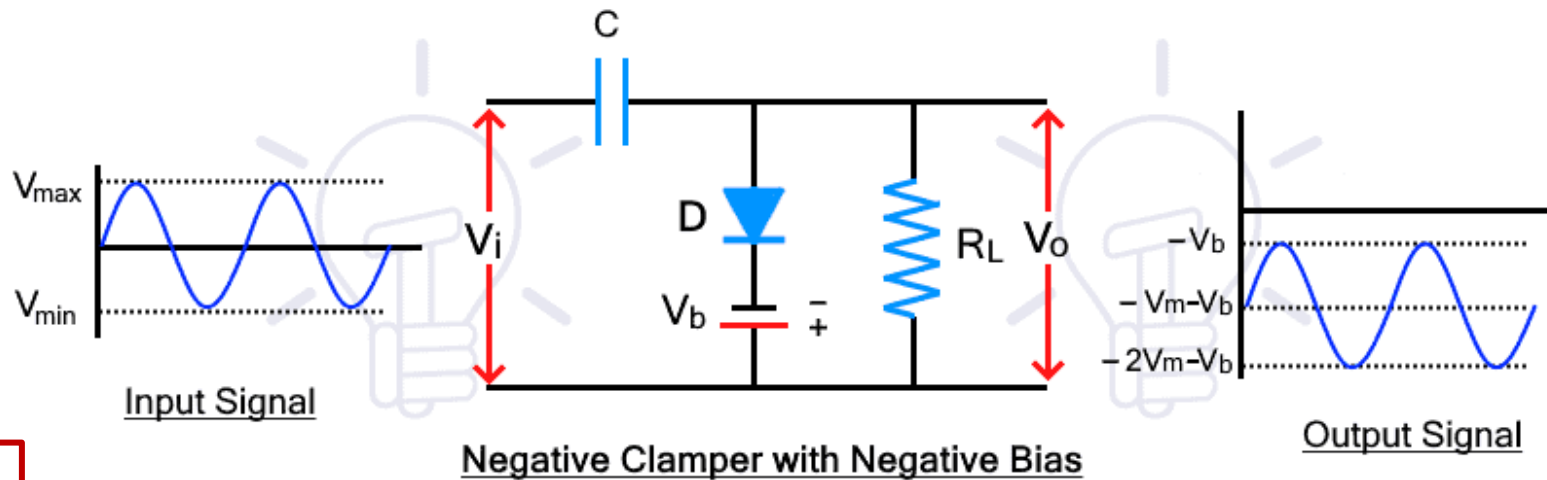
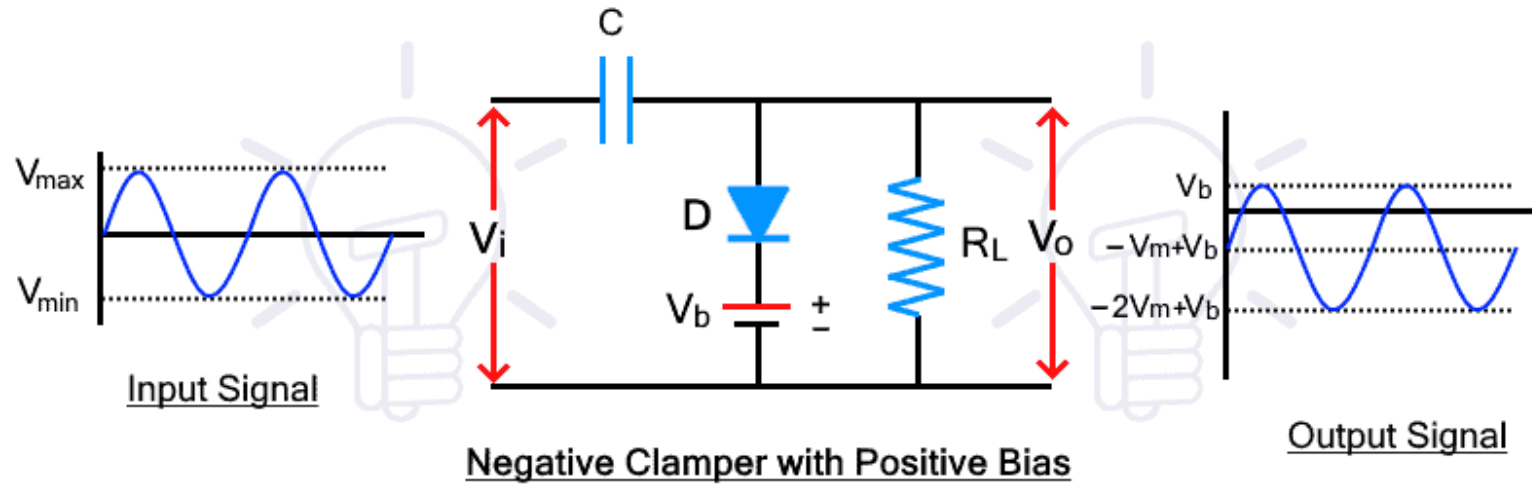
$$RC \gg T$$

Clamper Circuit



$$RC \gg T$$

Clamper Circuit



$$RC \gg T$$